Remote Sensing of Environment 333 (2026) 115164

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

Groundwater volume loss and land subsidence in the North China plain investigated using wide-area InSAR surveying and mechanical modeling☆

Xing Zhang [a], Jun Hu [a,b,*], Mahdi Motagh [c,d], Mingjia Li [e], Yuedong Wang [f], Qiuhong Yang [g], Guangli Su [h], Haigang Wang [i,j]

[a] School of Geosciences and Info-Physics, Central South University, Changsha 410083, Hunan, China
[b] Hunan Geological Disaster Monitoring, Early Warning and Emergency Rescue Engineering Technology Research Center, Changsha 410004, China
[c] Section of Remote Sensing and Geoinformatics, GFZ Helmholtz Centre for Geosciences, 14473 Potsdam, Germany
[d] Institute of Photogrammetry and GeoInformation, Leibniz University Hannover, Hannover, Germany
[e] Department of Earth and Space Sciences, Southern University of Science and Technology, Shenzhen 518055, China
[f] School of Land Science and Technology, China University of Geosciences, Beijing 100083, China
[g] Department of Earth and Atmospheric Sciences, University of Houston, Houston, TX, USA
[h] First Monitoring and Application Center, CEA, Tianjin 300180, China
[i] Hebei Cangzhou Groundwater and Land Subsidence National Observation and Research Station, Cangzhou 061000, China
[j] Schools of Resources Environment and Tourism, Capital Normal University, Beijing 100048, China

ARTICLE INFO

Edited by Jing M. Chen

Keywords:
North China plain
Wide-area multi-track network adjustment model
InSAR-VSM
Groundwater storage loss

ABSTRACT

The long-term over-extraction of groundwater in the North China Plain (NCP) has resulted in permanent loss of groundwater storage, leading to geo-disasters, including surface subsidence, ground fissures, and seawater intrusion. Through integrating wide-area InSAR survey data with a mechanical model, this study reveals the detailed spatiotemporal dynamics of aquifer system deformation and groundwater storage loss (GWSL) across the entire plain. We used a Sentinel-1 A SAR dataset from 2017 to 2023, covering two ascending orbits (T40 and T142) with eight frames, along with time-series InSAR technology and a wide-area multi-track network adjustment model that incorporates spatial constraints. This allowed us to obtain seamless, millimeter-level accuracy time-series deformation sequences for the NCP. Three severe subsidence areas were identified, with the maximum vertical velocity exceeding −150 mm/year and the cumulative subsidence reaching up to 1 m. Thereafter, the time-series InSAR results and groundwater level data were modeled using cross-wavelet analysis to quantify the elastic recovery capacity of the Quaternary aquifer system in the NCP (with the elastic skeletal storage coefficient S_{ke} ranging from 0.0002 to 0.021). This revealed significant differences in the aquifer response between the piedmont plain (mostly elastic deformation) and the flood plain (mostly plastic deformation). There was no elastic recovery in the subsidence funnel areas of the flood plain. Subsequently, InSAR observations were integrated with the volume strain model (InSAR-VSM) to construct a novel groundwater loss inversion model, yielding the first 2-km resolution dataset of GWSL for the NCP. The average annual groundwater loss from 2017 to 2023 was estimated to be -4.346×10^8 to -8.692×10^8 m^3/yr. Finally, we systematically investigated the spatiotemporal variation patterns of land subsidence over the past 63 years in the NCP, under the combined influence of human activities (such as over-extraction of groundwater, the South-to-North Water Diversion Project (SNWDP), and policies prohibiting groundwater extraction) and natural factors (such as extreme rainfall events and geological structures). The deformation shows a spatial migration characteristic from urban areas to agricultural land, with different deformation trends in the NCP. While urban deformation has stopped or reversed, agricultural areas continue to experience intensifying subsidence. This study characterizes the stress evolution features of the aquifer system at an ultra-high spatial resolution, providing crucial scientific support for adaptive groundwater management and the development of subsidence mitigation strategies in water-scarce sedimentary basins.

☆ Given their role as Associate Editor, Mahdi Motagh had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor.

* Corresponding author at: School of Geosciences and Info-Physics, Central South University, Changsha 410083, Hunan, China.
E-mail address: csuhujun@csu.edu.cn (J. Hu).

https://doi.org/10.1016/j.rse.2025.115164
Received 24 March 2025; Received in revised form 28 October 2025; Accepted 23 November 2025
Available online 2 December 2025
0034-4257/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

Land subsidence, which is among the most significant geological hazards globally, has occurred in over 150 countries or regions, particularly in areas influenced by human factors and excessive groundwater extraction (e.g., Hasan et al., 2023; Herrera-Garcia et al., 2021; Jasechko et al., 2024). This phenomenon is especially pronounced in water-scarce plain regions such as the Mexico Plain (Cigna and Tapete, 2021), San Joaquin Valley Plain (Neely et al., 2021), Central Iran Plain (Haghshenas Haghighi and Motagh, 2024), India (Garg et al., 2022), and Afghanistan (Kakar et al., 2025), where the frequency and impact are more significant. Over-extraction of groundwater, in conjunction with population growth, accelerating urbanization, and intensive agricultural development, has become one of the primary causes of ground subsidence (e.g., Ao et al., 2024; Motagh et al., 2024). It is well established that prolonged groundwater extraction not only leads to a significant loss of groundwater storage but also triggers subsidence, ground fissures, seawater intrusion, and other geological hazards (e.g., Motagh et al., 2017; Payne et al., 2025). Among these hazards, groundwater storage loss (GWSL) is a key indicator for assessing groundwater resources, and ground subsidence is the most direct geological response to excessive groundwater extraction (Long et al., 2020).

GWSL typically occurs in regions with intensive agricultural irrigation and rapid urbanization. Prolonged over-extraction causes a continuous decline in groundwater levels, leading to aquifer compaction and surface subsidence, forming a vicious cycle that exacerbates surface deformation. To accurately quantify this process, a multi-source data inversion system has been established. This process can be analyzed using Earth Observation (EO) data (Rodell et al., 2018) and geophysical/hydrological models (Long et al., 2020; Peng et al., 2022). It is worth noting that the existing methods still face three major technical bottlenecks, despite their remarkable advantages, as follows: 1) spatial resolution limitations:The Gravity Recovery and Climate Experiment (GRACE) satellite data have a low spatial resolution (typically >300 km), making it difficult to characterize groundwater changes at local scales(Rodell et al., 2018); 2) parameter uncertainty: hydrological models rely on numerous aquifer parameters (e.g., permeability and storage coefficient) that are difficult to obtain precisely; and 3) high monitoring costs: traditional groundwater well monitoring networks are expensive to deploy and unevenly distributed, which limits their detailed assessment. Interferometric synthetic aperture radar (InSAR) technology, with its high spatial resolution (up to meter-scale) and wide-area coverage, has provided an innovative solution for groundwater monitoring. Chaussard and Farr (2019) successfully inverted the annual groundwater loss in the San Joaquin Valley basin in California, United States, by integrating InSAR deformation data with hydrological models. This technology not only resolves the millimeter-scale elastic deformations caused by seasonal groundwater extraction, but also captures the spatiotemporal evolution of cumulative plastic subsidence. Based on InSAR deformation field inversion, groundwater storage change models can provide high-resolution dynamic assessments for regional water resource management.

As one of the fastest-subsiding regions in the world, the North China Plain (NCP) has formed a subsidence funnel area of over 48,500 km^2, due to prolonged groundwater over-extraction between 1971 and 2015, which has resulted in an irreversible loss of groundwater resources. In recent years, extreme and prolonged drought events triggered by climate change, coupled with the construction and operation of reservoirs in the Taihang Mountains, have further intensified the region's dependence on groundwater. Thorough monitoring and analysis of land subsidence in the NCP has yielded significant findings (e.g., Bai et al., 2022; Dong et al., 2023; Shi et al., 2020; Zhang et al., 2025).However, most studies have focused on specific areas, such as the urbanized regions of Beijing, Tianjin, and Cangzhou, for examining deformation and hydrogeological characteristics. Only limited research has addressed the spatiotemporal evolution of surface deformation across the broader NCP, with a comprehensive analysis of the entire plain and a quantitative assessment of groundwater storage being lacking. The existing studies have the following three major limitations, as follows: 1) in the 65 % of the plain area covered by agricultural irrigation, the ground subsidence monitoring network density is insufficient, resulting in a lack of coupling analysis and aquifer elastic deformation; 2) the hydrological and geological heterogeneity formed from the piedmont plain to the coastal plain, resulting from the sedimentary sequences between the Taihang Mountains and Bohai Bay, has not been fully considered, especially the permeability gradient variation of the aquifer structure in these regions; and 3) the existing research is largely confined to phenomenological descriptions, lacking the construction of quantitative models of the ground subsidence dynamics and aquifer system stress-strain responses across multiple spatiotemporal scales. Furthermore, although InSAR is a powerful tool, fusing multi-track data covering a vast region like the NCP to create a seamless deformation map with a unified datum poses significant methodological challenges. The existing mainstream strategies face inherent limitations. One common approach, i.e., stitching adjacent tracks using tie-points, struggles with the accumulation of long-wavelength errors along the strip, such as uncorrected orbital trends and unwrapping inconsistencies (Ge et al., 2010). Another approach, relying on external Global Navigation Satellite System (GNSS) data for calibration, is often hampered by the sparse and uneven distribution of GNSS stations (Kalia et al., 2017). Consequently, most of the traditional methods adopt a step-wise processing approach, which leads to error propagation and makes it difficult to obtain a high-precision, seamless deformation field for the entire NCP. Therefore, to overcome these limitations, the primary objective of this study is to conduct the first comprehensive and quantitative assessment of GWSL and its spatiotemporal dynamics across the entire NCP, at an unprecedented high resolution. To achieve this, we integrated a large Sentinel-1 A SAR dataset (2017–2023) with a wide-area network adjustment method and an InSAR-volumetric strain model (InSAR-VSM) framework. The specific goals were to: 1) generate a seamless, millimeter-level vertical deformation field for the whole of the NCP; 2) quantify the high-resolution GWSL and characterize the mechanical responses of the aquifer system (S_{ke}) in different hydrogeological zones (piedmont plain and flood plain); 3) systematically analyze the complex drivers of the observed subsidence, investigating the interplay between natural controlling factors (such as geological structures) and key anthropogenic activities (such as the South-to-North Water Diversion Project (SNWDP) and agricultural practices); and 4) place the recent deformation patterns into a multi-decadal historical context (1960–2023) to reveal the long-term spatiotemporal evolution of land subsidence and the shifting influence of its primary drivers. This study provides critical scientific support for sustainable groundwater management in this vital region.

2. Study area

The NCP encompasses five municipalities and provinces, encompassing Beijing, Tianjin, Hebei, Shandong, and Henan, which are renowned as China's economic, cultural, political, and transportation centers. Covering an extensive area of 300,000 km^2 (representing 3.1 % of China's total landmass) and housing a population of 339 million (constituting 24.2 % of the nation's total population), as of December 2019, this region demonstrates significant demographic and geographic importance within the country's overall landscape. The NCP is the largest and most populous plain in China.

Situated on thick, unconsolidated Quaternary deposits, the NCP is divided from west to east into three main hydrogeological units: the piedmont plain, the flood plain, and the coastal plain (Fig. 1a). The main aquifer systems are divided into two major aquifer groups: the shallow aquifers of the piedmont plain and the deep aquifers of the flood plain (Fig. 2a). Natural groundwater recharge primarily comes from precipitation, lateral runoff from the Taihang and Yan mountains, and seepage

from the lower Yellow River. Most of the NCP is characterized by a temperate and subtropical monsoon climate, with 560 mm average annual precipitation concentrated during the summer monsoon period from June to September, which is insufficient for effective recharge. Critically, the different hydrogeological units exhibit distinct mechanical responses to groundwater extraction: the piedmont plain is dominated by elastic deformation, possessing some capacity for storage recovery, whereas the flood plain is characterized by plastic deformation, where the aquifer compaction process is largely irreversible, leading to permanent loss of storage space and exacerbating land subsidence.

Agricultural irrigation is the primary use of groundwater, with the irrigation ratio in the NCP reaching up to 90 %.Since the 1970s, the continuous expansion of both cultivated area and water demand has led to a significant increase in agricultural water consumption. This long-term, large-scale extraction has caused a continuous and rapid decline in the water levels of the shallow aquifers. Furthermore, the water quality of these aquifers has deteriorated due to the infiltration of surface pollutants (e.g., pesticides and fertilizers), rendering them unsuitable for irrigation. Consequently, agricultural irrigation has been forced to shift from the depleted shallow aquifers to deeper groundwater aquifers. This over-extraction at various depths has triggered a severe overall decline in groundwater storage, leading to uneven drops in water levels and the formation of a large-scale groundwater depression cone.

To mitigate the groundwater depletion, the Chinese government has enacted a series of water resource management measures since 2012. It has implemented the South-to-North Water Diversion Project (SNWDP) to optimize water resource allocation. These strong interventions have shown initial success: over the past decade, a slowdown and even rebound of land subsidence have been observed in urban areas such as Beijing、Tianjin and Taiyuan(Su et al., 2023; Tang et al., 2022; Zhang et al., 2025). However, the impact of these policies is markedly uneven, with the subsidence problem showing a trend of shifting from urban to agricultural areas. Meanwhile, land-use changes and the shift from crops to economic forests have caused water resource consumption to increase by 126 % between 2013 and 2022, further exacerbating the groundwater depletion (Dong et al., 2024). This phenomenon is consistent with the findings regarding the evolution mechanism of groundwater storage anomalies (Zhao et al., 2024), as shown in Fig. 2b. The current situation of water resources and land subsidence is extremely complex and requires a more detailed assessment.

There are numerous faults/ground fissures in the NCP (Fig. 1b), the names of which are provided in Table S1. According to previous research, the orientation of some of these faults/ground fissures restricts the extension of the groundwater cone of depression and the expansion of the land subsidence bowl(Gong et al., 2018). These faults/ground fissures are primarily distributed along the margins of subsidence areas, appearing in weak geological strata and zones of significant differential settlement, such as the Longyao ground fissure (Peng et al., 2016). However, previous studies have mostly focused on the effects of single or limited faults/ground fissures and have not fully revealed the overall impact pattern across the NCP. Therefore, building upon previous localized studies, this study utilized a comprehensive, wide-area InSAR dataset to systematically investigate the interaction mechanism between regional land subsidence and these controlling fault systems.

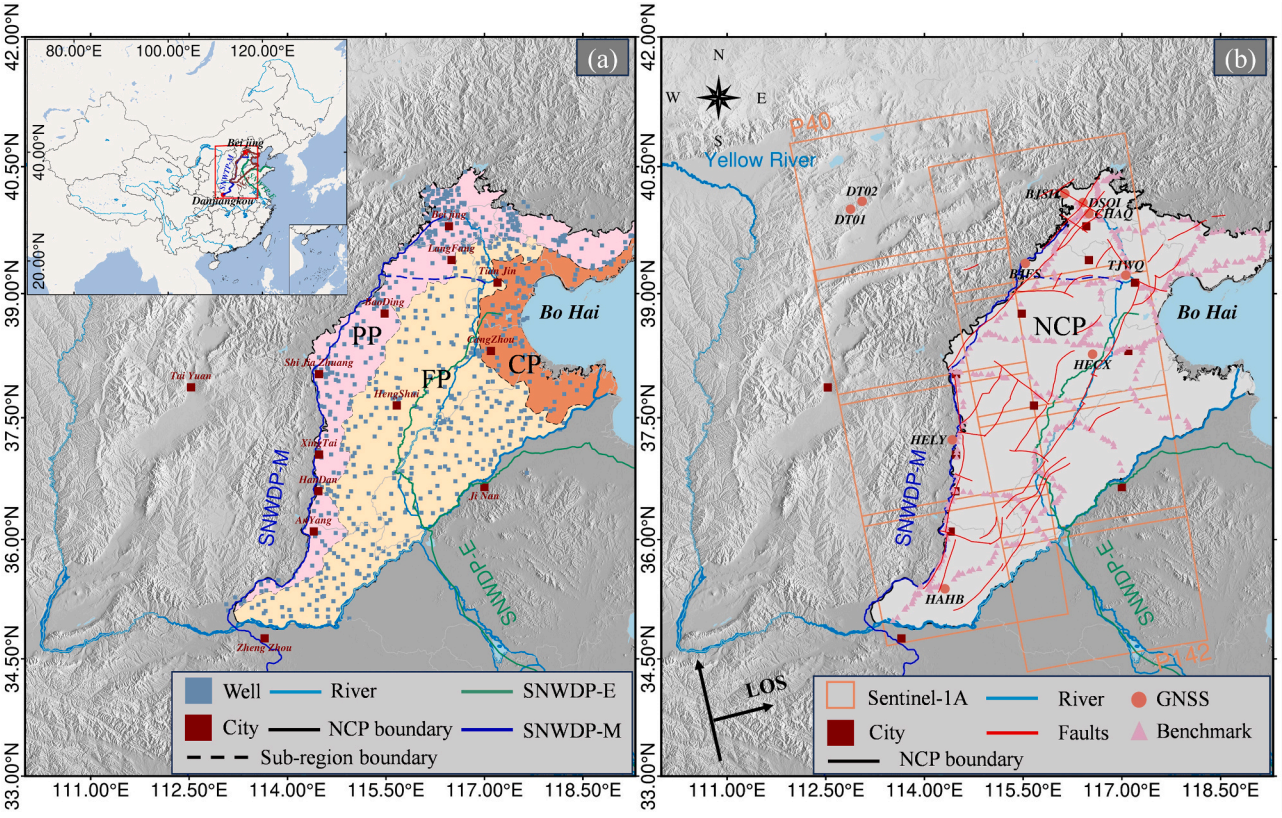

Fig. 1. (a) Location of the study region (orange: coastal plain (CP); yellow: flood plain (FP); pink: piedmont plain (PP)). The blue lines represent the middle route of the South-to-North Water Diversion Project (SNWDP-M); the green lines represent the eastern route of the South-to-North Water Diversion Project (SNWDP-E). The light blue solid squares represent the locations of the groundwater wells, and the dark red solid squares represent the major cities. (b) The orange rectangles represent the footprints of the used Sentinel-1 Interferometric Wide scenes; the pink triangles represent the benchmark points; the orange solid circles represent GNSS locations; and the red lines indicate the regional faults (Deng et al., 2003). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

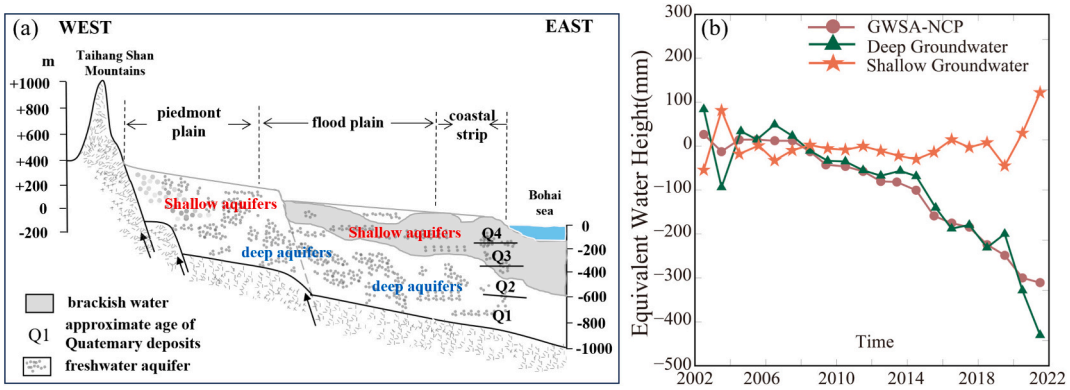

Fig. 2. (a) Hydrogeological conceptual model of the NCP Quaternary aquifer system, modified from Zhang et al. (2022). (b) Monthly groundwater storage anomalies (GWSA) and in situ observed trends in the NCP (2002–2021), modified from Zhao et al. (2024).

3. Geodetic and hydraulic datasets

3.1. InSAR data

In this study, eight stacks of 1500 Sentinel-1 A SAR scenes spanning from May 2017 to September 2023 were acquired in the ascending tracks T40 and T142. The multi-temporal SAR image stacks were systematically employed to monitor and analyze the land subsidence patterns. The acquisition parameters of the Sentinel-1 SAR datasets utilized in this study are comprehensively presented in Table 1. The geographical distribution and spatial extent of the SAR data stacks are shown in Fig. 1b.

3.2. GNSS data

The GNSS data employed in this study were obtained from three independent sources: the Crustal Movement Observation Network of China (CMONOC), the Beijing Continuous Operational Reference System (BJCORS), and the deployed GNSS stations. Specifically, the CMONOC dataset includes measurements from four CGPS stations (2011−2022), the BJCORS dataset contains measurements from two CGPS stations (2015–2019), and the GNSS observations also include measurements from two stations (2015–2023). All these GNSS stations are located within the study area (Fig. 1b). The data were first aligned with the ITRF2008 frame via a seven-parameter Helmert transformation (translation, rotation, and scaling), and subsequently corrected for residual errors before conversion to a Eurasian-fixed reference frame.

3.3. Leveling data

To decode the spatiotemporal evolution of surface subsidence in the NCP over a multidecadal scale, we integrated a unique legacy dataset comprising the following three epochs of second-order precision leveling surveys, conducted in accordance with the Chinese National Standard (GB/T 12897–2006): 1960–1980, 1980–1990, and 1990–2010. The 8200-km leveling network (Fig. 1b) strategically aligns with the major subsidence bowls and tectonic boundaries.

3.4. Groundwater level data

The groundwater level data used in this study were acquired from the China Institute of Geo-Environment Monitoring's Geo-Cloud platform (https://geocloud.cgs.gov.cn), comprising 15-day interval measurements from January 2018 to January 2023. Within the InSAR coverage area, 472 valid monitoring wells were identified (Fig. 1a). All the observation wells are located in confined aquifers and in flood plain areas, with depths ranging from 250 to 400 m. The land subsidence investigated in this study is primarily driven by the decline in piezometric head within these deep aquifers. Therefore, this study focused on the relationship between changes in the deep confined aquifers and land subsidence to explore future groundwater management strategies for the NCP.

4. Methodology

Figure 3 shows the main data processing workflow and research approach of this study. The research primarily consisted of two parts: 1)

Table 1
Parameters of the used SAR data.

Sensor	Orbit direction	Path	Frame	Time span	Wavelength (cm)	Imaging mode	Number of scenes	Total number of scenes
S1	Ascending	40	112	20170513–20230927	5.6	IW	188	1500
			117		5.6	IW	188	
			122		5.6	IW	188	
			127		5.6	IW	188	
		142	111	20170520–20230922	5.6	IW	187	
			116		5.6	IW	187	
			121		5.6	IW	187	
			126		5.6	IW	187	

determination of the deformation in the NCP; and 2) inversion of the groundwater parameters. Firstly, the multi-temporal interferometric synthetic aperture radar (MT-InSAR) technology was combined with the regional network adjustment method to derive comprehensive deformation measurements and to construct a spatiotemporally unified seamless vertical deformation field. Secondly, by developing a coupling model between the groundwater level changes and the time-series InSAR deformation, the elastic storage coefficient of the NCP was accurately calculated. Furthermore, based on the proposed InSAR-VSM model, quantitative assessment of the regional groundwater storage loss was achieved.

4.1. InSAR time-series processing

The MT-InSAR data processing over each SAR data frame followed the small baseline subset (SBAS) InSAR approach (Berardino et al., 2002). The entire workflow was conducted in distinct stages: the initial interferometric processing was performed with GAMMA software, while the subsequent time-series analysis was based on our own implementation of the SBAS algorithm in MATLAB. Firstly, we set the small spatial baseline B_p and temporal baseline $B_T(B_T = 80days, B_P = 250m)$ over each SAR data frame to select interferogram pairs, wherein the previous scene must form an interferogram pair with the next two scenes. The baseline diagram is shown in Fig.S 2. Next, precise image co-registration was conducted using the enhanced spectral diversity method to achieve an extremely high co-registration accuracy ($\sim 10^{-3}$ samples) (Yague-Martinez et al., 2016). Subsequently, the images were mosaicked into individual sub-swaths and the latter were integrated into a comprehensive mosaic covering the full spatial extent of the Sentinel-1 scene. We applied multi-looking factors of 20 by 4 for the range and azimuth directions, respectively, to effectively reduce phase noise in the interferometric phase data. The topographic phase component was eliminated from each interferogram using the AW3D30 digital elevation model (DEM) through differential interferometric processing. Following phase unwrapping processing with the extended minimum cost flow algorithm, the low coherence caused by crop growth in the NCP during the summer may result in unwrapping errors. Therefore, we further applied phase closure difference correction($\nabla\varnothing^{ij} + \nabla\varnothing^{jk} - \nabla\varnothing^{ik} = 0$) (Wang et al., 2023) to eliminate the impacts of unwrapping errors (Fig. S3). For interferograms covering areas with significant topographic relief, an elevation-dependent phase component, primarily attributed to tropospheric stratification, was removed using linear regression analysis. Subsequently, a quadratic polynomial model was applied to all the interferograms to remove the remaining phase ramp. Finally, the average velocity and time series at coherent targets along the line-of-sight (LOS) direction were retrieved via least-squares solutions.

4.2. Wide-area network adjustment algorithm

Basin-scale subsidence monitoring over the NCP necessitates multi-track and multi-frame SAR dataset integration. However, the following two fundamental challenges arise in wide-area InSAR mosaicking: 1) the incidence angle disparity within varied radar look angles (28° – 48° across adjacent tracks) induces systematic LOS deformation discrepancies in overlapping zones (Fig. 4a); and 2) reference frame inconsistency, where the relative deformation measurements from independent frames lack a unified geodetic datum due to arbitrary reference point selection. This benchmark difference was amplified during the wide-area stitching process, making it difficult to unify and accurately stitch the results. To address these challenges, we first solved the discrepancies in the LOS deformation within overlapping areas, which are caused by variations in the incidence angle. Specifically, we converted the satellite's LOS deformation to the vertical deformation ($UP_def = \frac{LOS}{COS(\theta)}$, where θ denotes the radar incidence angle) based on the core assumption that the horizontal deformation is negligible. In the NCP, where subsidence is primarily driven by groundwater extraction, which mainly results in vertical deformation, this assumption has been proven to be reasonable (Zhao et al., 2019). Through this step, we eliminated the influence of different viewing geometries. Secondly, we constructed a wide-area vertical deformation network adjustment model based on high-quality coherence points within overlapping regions of the SAR image datasets (Wang et al., 2024) (Fig. 4b). In Fig. 4b, the red triangles represent the unwrapping reference points in each frame (referred to as control points). The black plus signs represent the high-quality coherence points within the overlapping area of adjacent frames (referred to as high-quality corresponding points), only

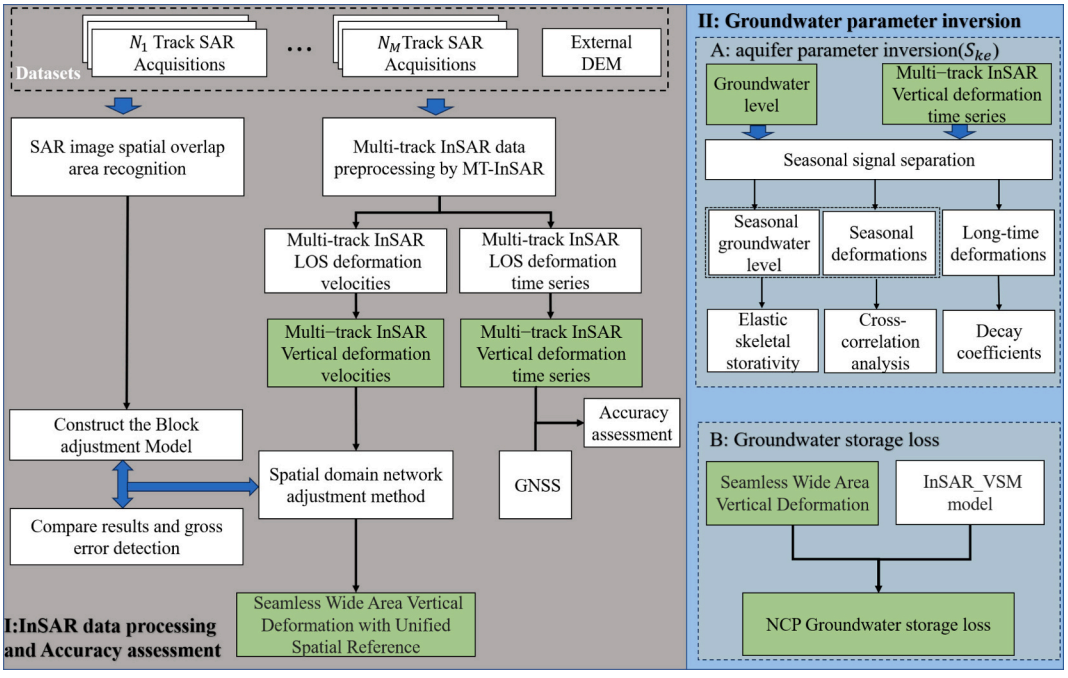

Fig. 3. Data processing and analysis flowchart.

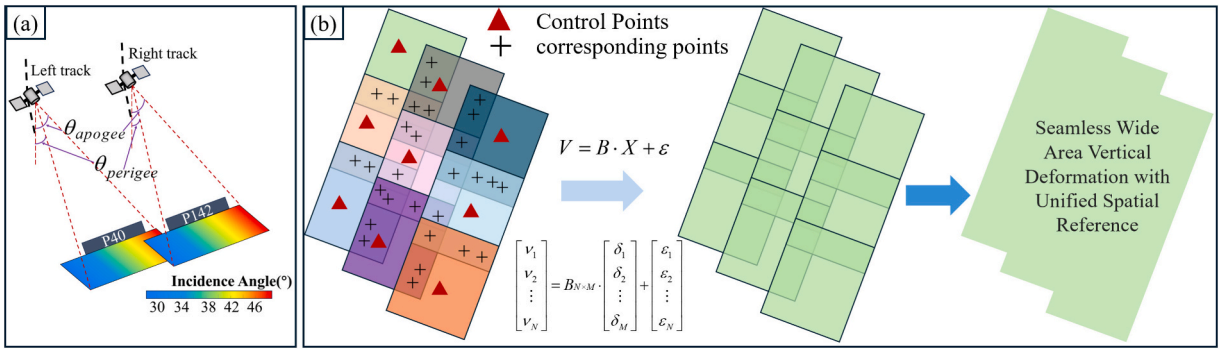

Fig. 4. (a) Schematic diagram of the difference in incidence angle between adjacent SAR satellites. (b) The regional network adjustment data processing. The red triangles represent the unwrapping reference points in each frame (referred to as control points), and the black plus signs represent the high-quality coherence points within the overlapping area of adjacent frames (referred to as corresponding points). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

considering corresponding points with an average temporal coherence greater than 0.85. Their characteristic high coherence ensures the reliability of the deformation measurements. By utilizing these high-quality coherence points in the overlapping regions, we could quantify and correct the discrepancies between different frames. More specifically, the selection of high-quality coherence points and their schematic representation can be found in the supplementary material (Figs. S6–8). In this study, we processed $M = 8$ independent vertical deformation rate maps (each SAR frame), which together formed $N = 13$ unique overlapping regions (see Fig. S8). We extracted the vertical deformation of the high-quality coherence points in the two images that made up the *ith* overlapping area and calculated their average values, $UP_def_i^1$ and $UP_def_i^2$, separately. In the absence of any errors, these two values should be equal ($UP_def_i^1 = UP_def_i^2$). Therefore, the objective function of this study aims to minimize the deviation between the observed values in all the overlapping areas within the entire region. The objective function is defined as follows:

$$\text{Objective Function} : min\left(\sum_{i=1}^{N}\left(\left|UP_def_i^1 - UP_def_i^2\right|\right)\right) \quad (1)$$

Eq. (1) serves as the objective function of an optimization problem, rather than a traditional equality relationship. It expresses the core optimization goal of the algorithm: by minimizing the discrepancies in the deformation observations in the overlapping regions between adjacent frames (both within the same track and across different tracks), their deformation measurements achieve optimal consistency in these areas. This minimization strategy provides the foundation for the subsequent mathematical adjustments to Eq. (2).

In this study, for N overlapping regions, a correction equation was constructed between the observation deviation of the overlapping regions and the frame correction amount, as follows:

$$\begin{bmatrix} v_1 \\ \vdots \\ v_N \end{bmatrix} = B_{N\times M} \bullet \begin{bmatrix} \delta_1 \\ \vdots \\ \delta_M \end{bmatrix} + \begin{bmatrix} \vartheta_1 \\ \vdots \\ \vartheta_N \end{bmatrix} \text{ or } V = B \bullet X + \varepsilon \quad (2)$$

where $V = [v_1 \ \cdots \ v_N]^T$ is the size, and $N \times 1$. $v_i = \left|UP_def_i^1 - UP_def_i^2\right|$ is the deviation value within each overlapping area. δ_i is the unknown parameter of $X = [\delta_1 \ \cdots \ \delta_M]^T$, with the size $M \times 1$, ϑ_i represents the residual, $\varepsilon = [\vartheta_1 \ \cdots \ \vartheta_N]^T$, and the size is $N \times 1$. B is the design coefficient matrix, and the size is $N \times M$. Each row represents an overlapping area, while each column represents a frame. The detailed construction of the B matrix, along with a concrete example (based on Figs. S6), was provided in the Supplementary Material. As a result of the InSAR data distribution and spatiotemporal decorrelation, the correction equations may suffer from rank deficiency. To enhance the robustness of the solution process, we adopted an equation-solving strategy based on the Moore-Penrose (MP) pseudoinverse algorithm, which was then used to obtain the InSAR observation correction values corresponding to each frame. After adjustment and correction, the vertical deformations of each frame exhibited high spatial consistency. To construct a wide-area seamless vertical deformation field for the NCP, it was necessary to fuse the multi-frame deformation results using data mosaicking techniques. The corresponding points in the overlapping regions were averaged. Finally, the mosaicked vertical deformation results were unified to a single stable reference benchmark, thereby obtaining a wide-area seamless deformation field with a spatiotemporally unified datum for the NCP.

4.3. Aquifer parameter estimation

We adopted the single-sinusoid model to extract the long-term and seasonal trends of the time-series InSAR deformation and groundwater level data. The purpose was to quantify the elastic release coefficient, which can be mathematically expressed as follows:

$$Y_1(t_1) = v_1 t_1 + A_1 cos\left(2\pi(t_1 - T_1)\right) + B_1 \quad (3)$$

$$Y_2(t_2) = v_2 t_2 + A_2 cos\left(2\pi(t_2 - T_2)\right) + B_2 \quad (4)$$

where $Y_1(t_1)$ and $Y_2(t_2)$ represent the time series of InSAR deformation and groundwater level data, respectively.

The other parameters are defined as follows:

v_1 and v_2 denote the long-term linear trend rate of the InSAR deformation and groundwater level data;

A_1 and A_2 are the amplitudes of the seasonal InSAR deformation and groundwater level data;

t_1 and $t_2$are the time variables for the InSAR deformation and groundwater level series;

T_1 and T_2 indicate the peak timing of the seasonal signals of the InSAR deformation and groundwater level data (where $T = 0$ corresponds to October 1);

B_1 and B_2 are constant terms.

The change in groundwater storage status refers to the change in water storage capacity between the weakly permeable layers (clay and silt) and their underlying aquifers (gravel). This is represented by the elastic skeletal storage coefficient (S_{ke}), which is defined as the volume of water released from or absorbed by aquifer storage per unit area in response to a unit change in groundwater level. The properties of an aquifer system can be quantitatively determined by analyzing the temporal variations in both its thickness and the corresponding groundwater level fluctuations. A key challenge, however, is the difficulty in obtaining the initial groundwater level, which hinders the accuracy of these calculations. This can be achieved through the elastic deformation

and the corresponding groundwater level change using the following formula (Smith and Li, 2021):

$$S_{ke} = \frac{\Delta d_s}{\Delta h_s} \tag{5}$$

where Δd_s denotes the amplitude of the seasonal aquifer thickness variations, corresponding to the seasonal deformation, andΔh_s denotes the seasonal variation of the groundwater level.

4.4. Groundwater volume loss assessment

Changes in groundwater volume are among the most important parameters in groundwater inversion. These values can be used to understand groundwater activities and evaluate disaster severity caused by groundwater over-exploitation. Zhang et al. (2024) proposed a method incorporating InSAR measurements with a volume strain model (VSM), enabling the estimation of high-resolution sub-surface fluid volumetric changes (InSAR-VSM). The fundamental principle of the InSAR-VSM model is rooted in the elastic half-space theory. It establishes a direct physical relationship, where the observed surface deformation is caused by the volumetric compaction or expansion within the subsurface aquifer system due to fluid pressure changes, and can be expressed as shown in Eq. (6). Rather than making prior assumptions about the specific geometric shape of the groundwater reservoir, the model discretizes the aquifer space into numerous homogeneous units, such as parallelepiped or columnar cells.

$$L(x) = \int_v G(x,y)dv(y)dy \tag{6}$$

where $L(x)$ represents the surface displacement at point x, $dv(y)$ represents the subsurface fluid volumetric change of block y with a source volume V, and G is the Green's function. The Green's function represents the vertical displacement response at the surface caused by a unit volumetric strain within an aquifer cell, assuming a homogeneous, isotropic, elastic half-space. It serves as a physical model connecting the subsurface processes to the observed surface deformation. For the calculation, a Poisson's ratio of 0.25 was assumed.

In this study, we divided the groundwater reservoir space into m blocks dv_i ($i = 1, 2, \ldots, m$). Based on the InSAR-VSM model, the following formula was constructed for the vertical deformation of n InSAR observations $L_j (j = 1, 2, \ldots n)$ and the m unknown volume change for each block. The relationship is formulated as:

$$\begin{bmatrix} L_1 \\ \vdots \\ L_n \end{bmatrix} = \begin{bmatrix} G_{11} & \cdots & G_{1m} \\ \vdots & \ddots & \vdots \\ G_{n1} & \cdots & G_{nm} \end{bmatrix} \begin{bmatrix} dv_1 \\ \vdots \\ dv_m \end{bmatrix} + \begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix}, P = D_{LL}^{-1} \tag{7}$$

where n and m are the numbers of observations and blocks, respectively; r_j is the observation residual; D_{LL} is the observation variance-covariance matrix, assumed to be a diagonal matrix, reflecting the assumption that InSAR observations are statistically independent,$D_{LL} = (\sigma_1^2, \sigma_2^2, \cdots, \sigma_n^2)$. The variance σ_i^2 for j th observation was estimated based on its average interferometric coherence δ_i^2 using the relationship($\sigma^2 \propto 1/\delta^2$); and G_{ji} is the Green's function, representing the relationship between the volumetric strain variations in the ith source block and the corresponding vertical surface displacement at the jth observation point. We set $L = [L_1 \quad \cdots \quad L_n]^T$, $X = [dv_1 \quad \cdots \quad dv_m]^T$,$B = \begin{bmatrix} G_{11} & \cdots & G_{1m} \\ \vdots & \ddots & \vdots \\ G_{n1} & \cdots & G_{nm} \end{bmatrix}$, and $l = [r_1 \quad \cdots \quad r_n]^T$. Eq. (7) can then be simplified as:

$$L = BX + l \tag{8}$$

We minimized the second parameter gradients to obtain a smooth solution that fulfills the least-squares error criteria. Therefore, the Eq. (8) subject was solved to the following condition:

$$\|Pl\|^2 + \gamma^2 \|DX\|^2 \rightarrow min \tag{9}$$

where D is the Laplace operator, implemented as a finite-difference approximation of the second spatial derivatives to enforce smoothness on X; the operator $||.||$ represents the L_2 norm, and γ is a smoothing factor. The optimal value for γ was selected using the L-curve method (Aster et al., 2018)to balance data misfit and solution roughness in the least-squares adjustment for solving Eqs. (8) and (9).

Finally, the GWSL is obtained by summing the volume changes from all the individual cells, incorporating the porosity (n) of the aquifer. The equationGWSL $= nh\, l_c w_c \sum d_v$ directly demonstrates how the calculated volumetric strain (d_v) for each cell (with dimensions h, l_c, and w_c) is converted into groundwater volume change using porosity.

5. Results

5.1. Land subsidence patterns and rates

Two LOS subsidence rate maps were generated from the ascending P40 and P142 orbit data from May 2017 to September 2023 (Fig. 5). The InSAR results from the P40 orbit mainly cover the piedmont plain areas of the Taihang Mountains and parts of Shanxi, revealing widespread ground subsidence (Fig. 5a). Although deformation data in the mining areas are scattered, the deformation magnitude is major. The P142 orbit InSAR results primarily involved the flood plain and large urban areas such as Beijing and Tianjin, where ground subsidence and localized subsidence regions coexist (Fig. 5b). Based on these rate maps, we found that the maximum subsidence rate in the subsidence funnels for both the P40 and P142 orbits exceeds −120 mm/year, and the maximum subsidence rate for the other localized subsidence funnels exceeds −100 mm/year. Highlighting the spatiotemporal distribution of the subsidence funnels, Fig. 6a–d shows four subsidence profiles (i.e., A–A', B–B′, C–C′, and D–D′ from Fig. 5). From the profile analysis, it is evident that the A–A' profile contains six independent subsidence funnels, while the B–B′ and D–D′ profiles each contain two independent subsidence funnels, and the C–C′ profile (approximately 200-km long) contains three larger subsidence funnels. Over time, these subsidence funnels will continue to expand, potentially forming a super-large subsidence funnel. To better illustrate the temporal distribution of the deformation at the center of the subsidence funnels, Fig. 7a–c shows the deformation time series for points P1–P3. The results show that the subsidence at points P1–P3 exhibits a linear trend with periodic variations. The deformation time series for points P1–P3 reveal that subsidence occurs from March to September each year, with a subsidence reduction period from October to the following February. This pattern aligns with the seasonal agricultural irrigation in the NCP region, which drives groundwater extraction and recharge.

5.2. Regional network adjustment results

The distinct observation geometries between the P40 and P142 tracks, marked by an incidence angle difference of nearly 20°, introduce significant, systematic discrepancies into their LOS deformation measurements, even within overlapping areas (Fig. 4). While the visual patterns of subsidence appear broadly similar (Fig. 5), quantitative analysis reveals critical biases. For instance, the deformation rates at co-located points P2 and P2' exhibit a relative difference of approximately 10 % (Fig. 7). Consequently, a simplistic merging technique that only adjusts for a mean offset is insufficient, as it cannot correct for spatially varying, angle-dependent distortion and would inevitably leave prominent step-like artifacts at the track boundaries. To address this challenge and construct a seamless, wide-area vertical deformation field, we implemented the regional network adjustment detailed in Section 4.2. The efficacy of this method is rigorously demonstrated in Fig. 8. The adjustment transforms the statistical distribution of the inter-track

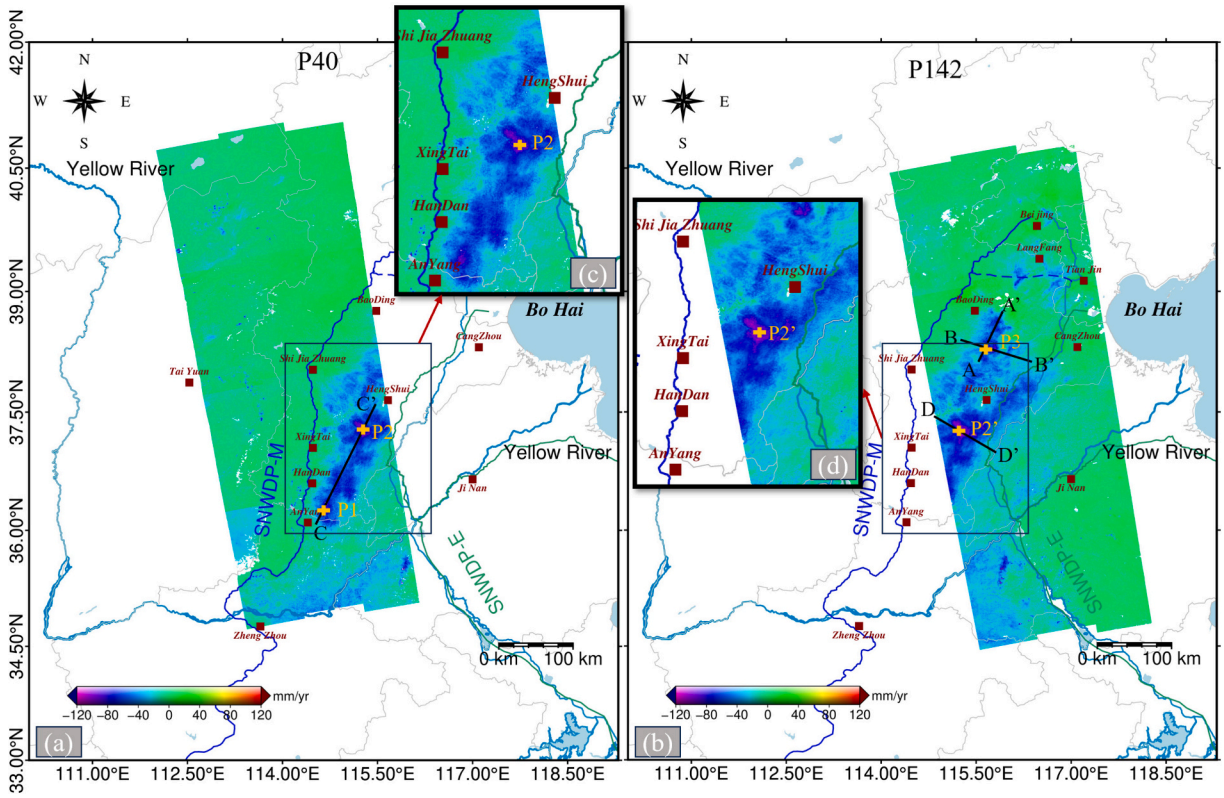

Fig. 5. The average deformation rates in the LOS direction over the NCP, spanning from 2017 to 2023. (a) LOS deformation rates from Sentinel-1 Path 40 (P40). (b) LOS deformation rates from Sentinel-1 Path 142 (P142). (c) Enlarged view of the deformation rates from Path 40 for the area indicated by the black solid rectangle in (a). (d) Enlarged view of the deformation rates from Path 142 for the corresponding area indicated by the black solid rectangle in (b), which is the same region shown in (c).

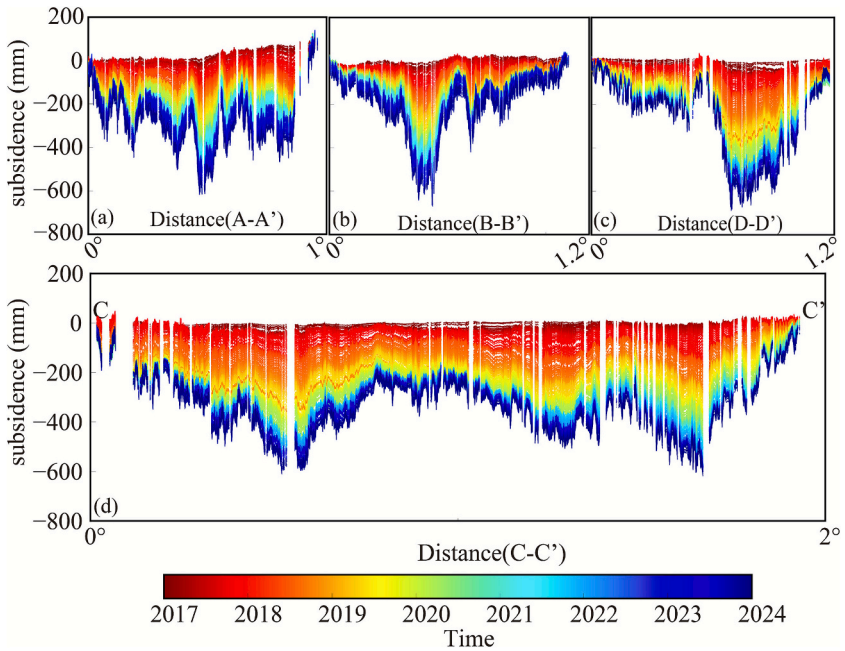

Fig. 6. (a)–(d) The temporal evolution of subsidence along the profiles (the black dotted lines in Fig. 5).

differences from a diffuse, bimodal histogram with a mean bias of −4.3 mm/yr (Fig. 8e) to a concentrated, single-peak, near-Gaussian distribution, with its mean reduced to −2.1 mm/yr (Fig. 8f). Furthermore, the profile line across the track boundary (Fig. 8c) provides clear evidence of enhanced spatial continuity, showing the complete removal of the abrupt "step" artifact present in the preliminary data. These results provide unequivocal validation that the proposed network adjustment procedure is both a necessary and highly effective methodology for fusing multi-frame InSAR data.

With its high fidelity and continuity thus validated, the resulting

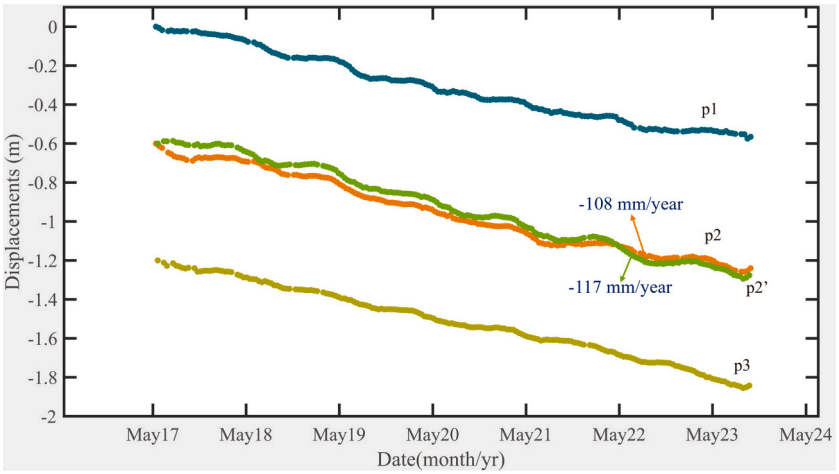

Fig. 7. The cumulative subsidence over time for P1 to P3.

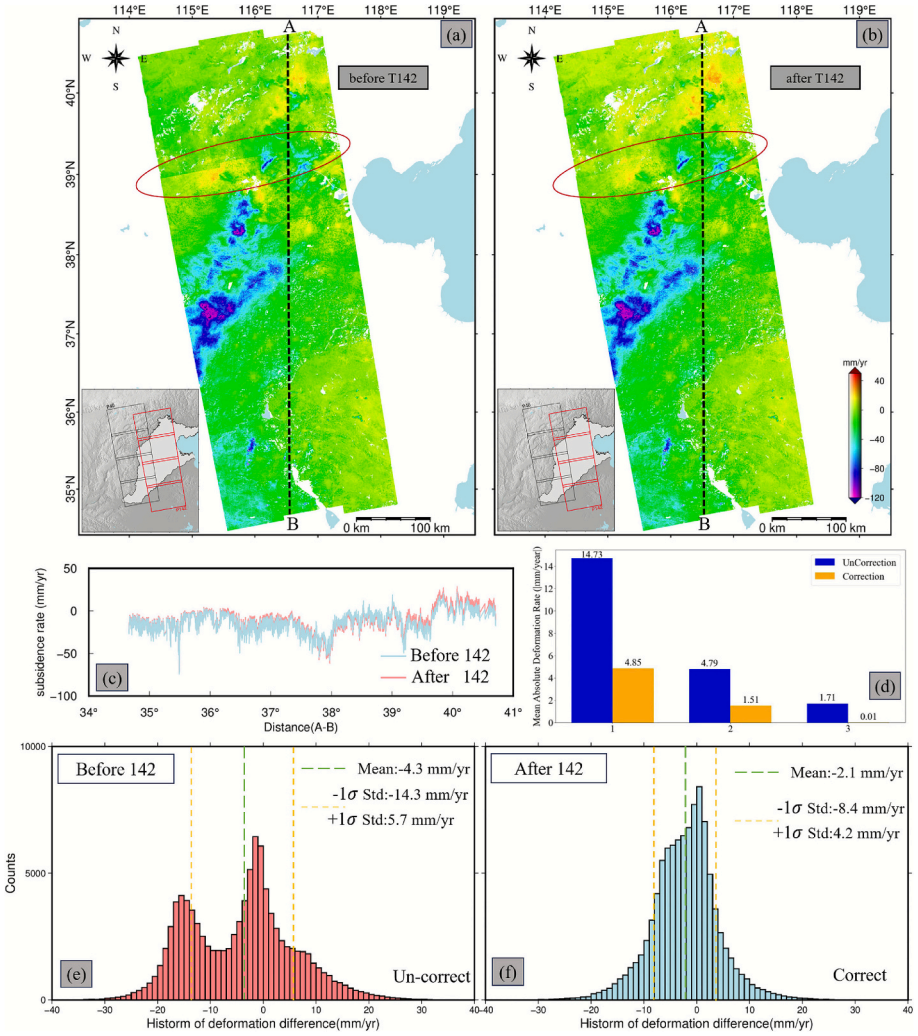

Fig. 8. Validation of the regional network adjustment for fusing multi-track data. (a), (b) Subsidence rate maps before and after adjustment. (c) Profile along the A–B transect, showing the removal of a step-like discontinuity at the track boundary. (d) Mean absolute rate differences in the overlapping area before (blue) and after (orange) correction. (e), (f) Histograms of therate differences (or errors) before and after correction, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

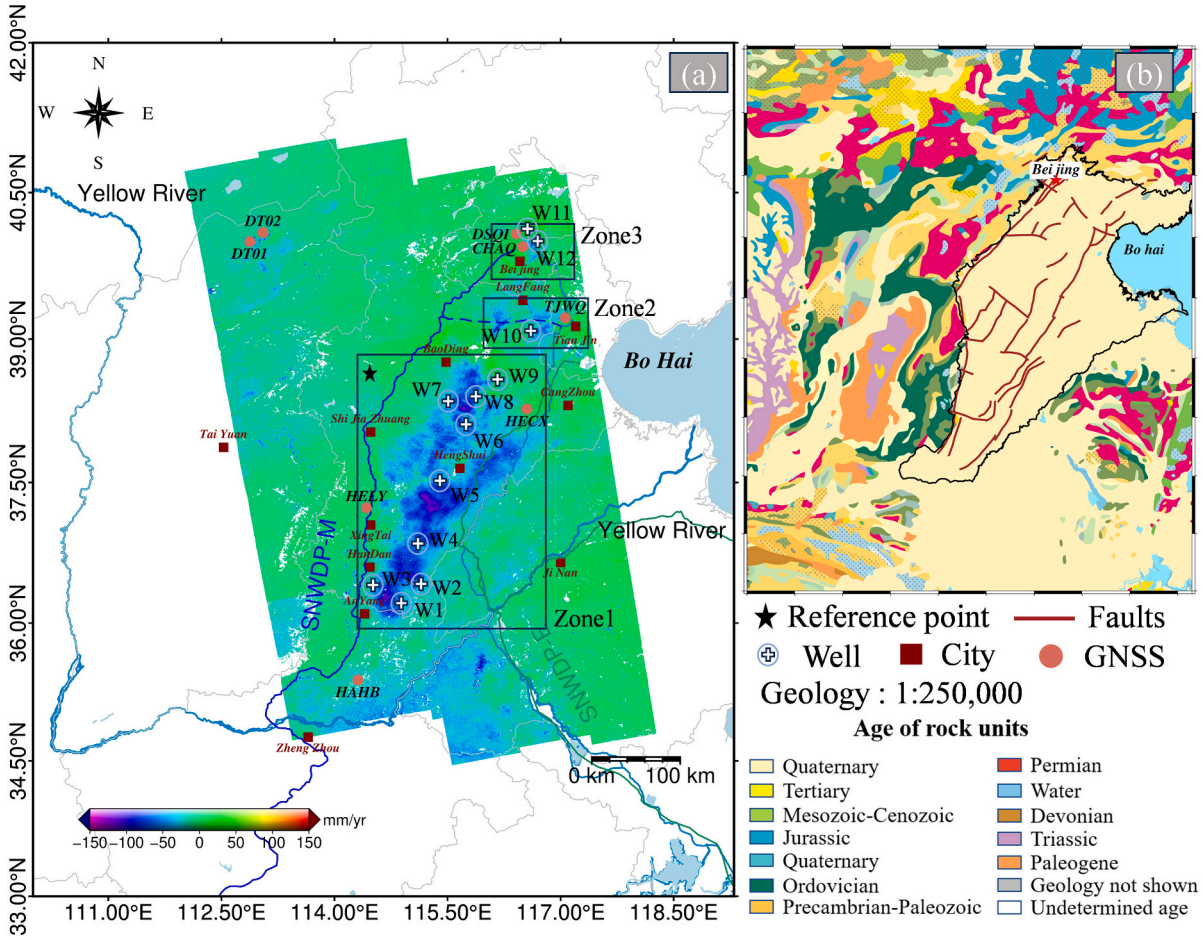

Fig. 9. (a) The vertical deformation result after regional network adjustment. (b) Geological map and 1:5,000,000 scale geology in the study area (Steinshouer et al., 1997). Zones 1, 2, and 3 represent the three main subsidence funnel areas located in the main agricultural irrigation area of the NCP, the suburbs of Tianjin, and the suburbs of Beijing, respectively.

unified vertical deformation field (Fig. 9) offers a comprehensive view of land subsidence across the NCP. The three main subsidence funnel zones in the NCP (Zones 1–3) are concentrated in a narrow, elongated band extending from southwest to northeast (Fig. 9a). These deformation areas are all situated within regions covered by thick, unconsolidated Quaternary deposits (Fig. 9b). These deposits are composed of multiple alternating layers of sand, silt, and clay, and the subsidence is primarily caused by the compaction of the compressible clayey layers following extensive groundwater extraction. Zone 1, which is approximately 200 km by 500 km, is a large agricultural irrigation area with a continuous large subsidence funnel, the largest in the world. Zones 2 and 3 are located in the suburbs of Tianjin and Beijing, respectively, which are also areas with intense groundwater extraction. In the NCP, the available surface runoff and natural groundwater recharge are insufficient to meet the water demands of agricultural irrigation and urban use, leading to heavy reliance on deep groundwater extraction. As the distance from the Taihang Mountains and the major surface runoff areas increases, the ground subsidence phenomenon is becoming more pronounced. This is due to the reduced intensity of groundwater recharge as the distance from the recharge areas increases, resulting in a situation where the amount of groundwater recharge is far smaller than the amount of extraction, leading to net losses in the aquifer and increased crustal stress, thereby forming dense subsidence funnel areas.

5.3. Land subsidence in response to groundwater levels

To reveal the relationship between land deformation and groundwater levels, we collected time-series data from 300 groundwater wells. This complete dataset was utilized for the spatial analysis of aquifer properties, as described in Section 5.4. For the detailed temporal analysis presented in this section, we selected 12 typical wells to illustrate the coupling relationship. Fig. 10 shows the residual components for the 12 selected typical well locations (Fig. 9a), which were obtained by removing the linear trends of both the groundwater level and vertical deformation from the original time series (Fig. S9), revealing a strong periodic signal. The depths of these 12 wells, all of which exceed 178 m, are provided in Table 2. The InSAR time-series trend aligns closely with the groundwater level trend, demonstrating a significant positive correlation ($\geq$0.94) between the land subsidence and the decrease in groundwater level (Fig. 10). Groundwater in the NCP is extensively used for agricultural irrigation, with an average extraction depth of more than 200 m, especially in flood plain areas, where the average groundwater level depth exceeds 300 m. This indicates that land subsidence in the NCP is primarily driven by extensive extraction from deep aquifer systems, while the compaction of sedimentary layers results from the depletion of groundwater resources. In addition, we performed cross-wavelet analysis (XWT) between the InSAR-derived deformation and groundwater level data from the typical wells. The results for well W8 (Fig. 11) reveal a strong relationship between the two signals, dominated by a primary resonance period of one year, which indicates a clear seasonal cycle. Specifically, the cross-wavelet spectrum shows a region of high spectral power (the warm yellow colour), statistically significant at the 95 % confidence level (the thick black contour), centered on the annual period from 2019 to 2021. Within this region, the phase arrows

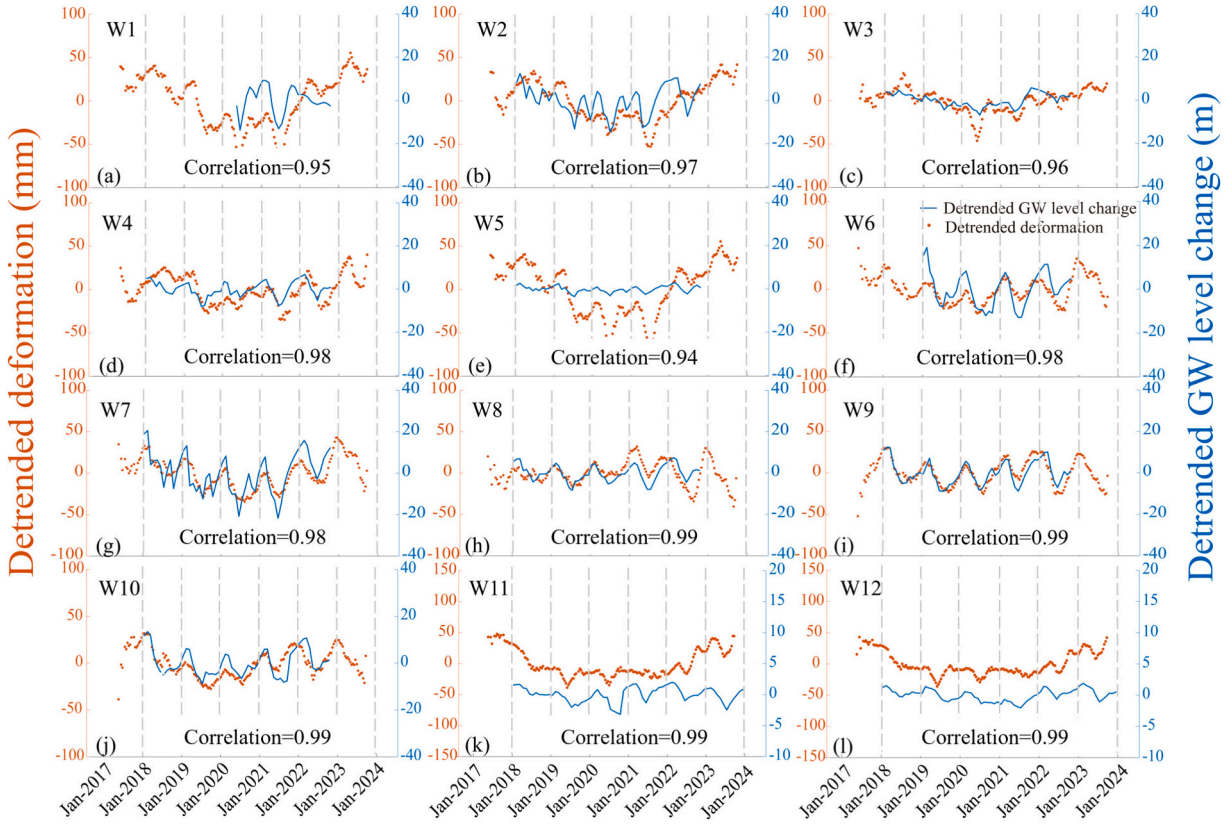

Fig. 10. Comparison of the detrended time-series InSAR deformation and groundwater levels for the 12 typical well locations shown in Fig. 9a.

Table 2
Depths of the 12 selected groundwater wells.

Groundwater well ID	Depth (m)	Groundwater well ID	Depth (m)
W 1	204.4	W 7	290.5
W 2	210.4	W 8	348.5
W 3	276.5	W 9	336.7
W 4	330	W 10	378
W 5	315.5	W 11	276
W 6	317	W 12	178.9

consistently point to the right direction, which indicates that the seasonal surface deformation and groundwater levels are strongly in-phase, with the seasonal deformation lagging the seasonal groundwater level by 2 days ($\nabla\theta = 2^{\circ}$). This short time delay suggests that surface deformation responds very quickly to groundwater level changes. The cross-wavelet analysis provides a more detailed understanding of the relationship between aquifer system changes and deformation, supporting our analysis and monitoring of groundwater variations in the NCP.

5.4. Skeletal storage coefficient estimation

Based on the deformation and groundwater level modeling method presented in Section 4.2, we performed point-by-point modeling of the vertical temporal deformation across the NCP. The periodic deformation and long-term deformation were extracted (Fig. S10) and the seasonal groundwater deformation was obtained (Fig. 12a). In addition, we extracted the seasonal deformation from InSAR data near the groundwater well locations and identified the times of the maximum seasonal amplitude (Fig. 12 b–c). Using Eq. (5), we plotted the distribution of groundwater S_{ke} across the entire NCP (Fig. 12d).

As shown in Fig. 12a and c, a strong positive correlation exists between the seasonal surface deformation and the groundwater

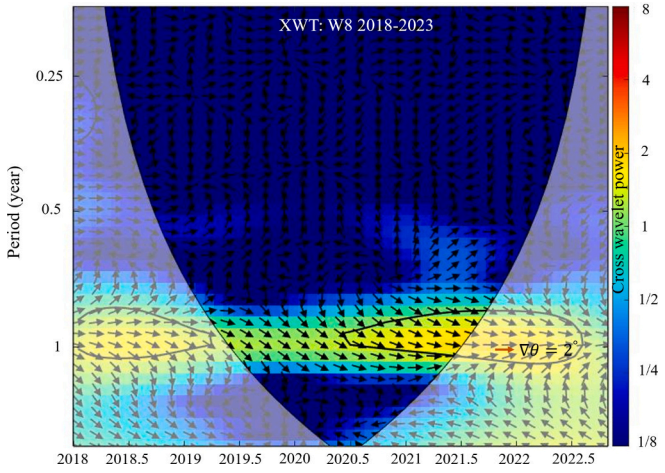

Fig. 11. Cross-wavelet transform (XWT) analysis of the seasonal groundwater level at the W8 well and seasonal surface deformation from 2018 to 2023. The horizontal axis shows the time, and the vertical axis shows the period in years. The colour represents the cross-wavelet power, with the warm colour (yellow) representing a high common power (strong correlation) and the cool colour (blue) representing a low common power. The arrows indicate the phase relationship between the two series, and arrows pointing to the right signify an in-phase (positive) relationship.The black contour indicates the 95 % significance level. The shaded region is the cone of influence (COI), where edge effects make the results unreliable. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

fluctuation in the floodplain areas. In areas where the groundwater fluctuation amplitudes are as high as 8–16 m, the surface also undergoes substantial seasonal deformation. This phenomenon reflects the area's low skeletal storage coefficient (predominantly <0.01, Fig. 12d). This

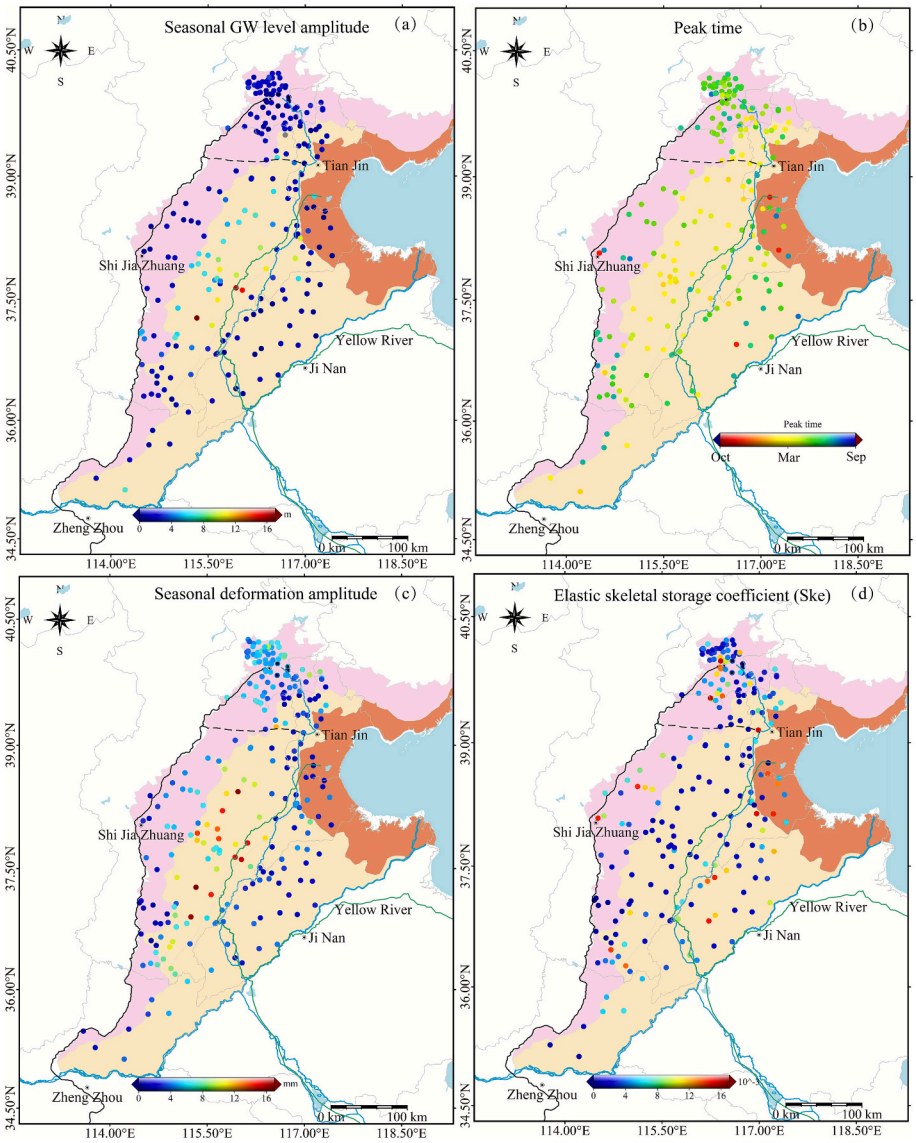

Fig. 12. Groundwater parameters of the NCP. (a) Seasonal groundwater level amplitude. (b) Timing of the seasonal level peaks. (c) Seasonal InSAR deformation amplitude. (d) Elastic groundwater storage coefficient (S_{ke}).

low coefficient is a direct measure of the aquifer system's poor elastic recovery capacity and provides a quantitative confirmation of the dominance of plastic deformation in this region. This indicates that, for every meter of groundwater level recovery, the ground surface rebounds only minimally, signifying that much of the prior compaction is irreversible. This is in stark contrast to the piedmont plain, where higherS_{ke} values suggest a more elastic response to head fluctuations. Consequently, the sustained over-extraction in the floodplain, coupled with this low elastic storage, leads to compounding, permanent surface compaction and a severe, long-term loss of aquifer storage. Under natural conditions, precipitation in the NCP is predominantly concentrated between July and September annually, with groundwater levels generally reaching their peak from August to October. The groundwater trend is consistent with the phenomenon observed in the piedmont plain region, as precipitation effectively replenishes the aquifers in the piedmont plain (Fig. 12b). However, the seasonal variations in the floodplain areas differ, with the peak of the seasonal surface deformation in the floodplain occurring from January to March, aligning with the seasonal fluctuations of the head changes (Figs. 10 and 12b). The groundwater level in the floodplain is primarily influenced by agricultural irrigation, with the highest levels occurring from January to March and the lowest levels from June to July. This indicates that surface subsidence in the floodplain is mainly caused by excessive groundwater extraction.

5.5. Estimating groundwater storage volume loss

Eqs. (8) and (9) were utilized to estimate the variations in water storage and to quantify the stress contributions resulting from the deformation of the aquifer system. We approximated the multi-layer groundwater aquifer by a thicker single-layer groundwater aquifer(Hu and Bürgmann, 2020) and applied the VSM to the agricultural irrigation area (Zone 1), as shown in Fig. 9.

We initially performed downsampling processing on the subsidence monitoring data from Zone 1 to achieve a uniform 1 km × 1 km resolution. To address the geometric irregularity of the groundwater reservoirs, a discretized grid system was established with 100 cells in the east–west direction and 150 cells in the north–south direction, yielding a total dimension of 200 km × 300 km (cell resolution: 2 km × 2 km). The analysis was based on data from the deep confined aquifer system. As stated in Section 3.4, the monitoring wells in the dataset used in this study have depths ranging from 250 to 400 m, indicating that drawdown from groundwater extraction affects the aquifer system to at least this

depth. We conceptualized the aquifer system as a continuous-medium reservoir with 500-m thickness in the InSAR-VSM coupled model. Based on lithological testing of the Quaternary unconsolidated strata in the study area, a Poisson's ratio of 0.25 was adopted to characterize the lateral deformation response of the aquifer medium. Based on the proposed model, the volumetric strain distribution of Zone 1 was obtained (Fig. 13), with the maximum volumetric strain (D_{vmax}) reaching $-9.4*10^{-8}$. To further validate the accuracy of the calculated volumetric strain, we conducted forward modeling verification (Fig. 14). The residuals exhibit a random distribution pattern, with a maximum residual of 20 mm, demonstrating the accuracy of the derived volumetric strain results (Fig. 14c). The groundwater volume loss is presented as $GWSL = nh\, l_c w_c \sum d_v,\, l_c$, where w_c represents 2-km cell dimensions and h represents 500-m cell dimensions, with the porosity n ranging from 0.2 to 0.4. Within Zone 1, which constitutes the primary agricultural irrigation area of the NCP, the calculated annual groundwater volume change reaches -4.346×10^8 to -8.692×10^8 m^3/yr. Given that this specific zone's groundwater use is overwhelmingly for agriculture, these results highlight the immense scale of the groundwater loss largely driven by irrigation in this agricultural region of the NCP from 2017 to 2023. Spatial heterogeneity analysis reveals the maximum water loss to be concentrated in the core subsidence funnel area, whose synergistic effects ultimately formed the large-scale continuous subsidence funnel.

6. Discussion

6.1. Reliability assessment and methodological limitations

To evaluate the reliability of this study, we conducted a multi-stage assessment. Firstly, we performed an accuracy analysis by comparing the InSAR results with the GNSS monitoring results. Specifically, we compared the obtained time-series InSAR results with the vertical deformation data from eight GNSS monitoring points located in the subsidence funnels (Figs. 1b and 9a) to verify the accuracy of the vertical deformation. This comparison demonstrated a robust agreement, with the root-mean-square error (RMSE) ranging from a minimum of 4.8 mm to a maximum of 13.4 mm. Furthermore, the analysis reveals a high degree of temporal correlation, evidenced by R^2 values ranging from 0.94 to 0.98 (Fig. 15). Based on these validation results, we conclude that the InSAR methodology employed in this study exhibits high accuracy and consistency with the independent GNSS monitoring data, thereby indicating its capability to reliably capture the dynamic changes in large-scale vertical deformation.Moreover, we further assessed the robustness of the final GWSL estimates through cross-validation with a widely used estimation method based on one-dimensional consolidation theory. This approach is based on the principle that long-term subsidence in heavily exploited confined aquifers, such as those in the NCP, is predominantly driven by inelastic deformation and directly corresponds to permanent storage loss. Following this method, we used the mean vertical subsidence rate to estimate the mean annual GWSL, yielding a result of approximately -1.26×10^9 m^3/yr. Although this value is numerically different from our InSAR-VSM result ($-4.346 \sim -8.692 \times 10^8$ m^3/yr), they are comparable in order of magnitude. We interpret this not as a discrepancy, but as strong cross-validation. Despite the different physical underpinnings of the two methods (three-dimensional physical inversion vs. one-dimensional consolidation theory), they both point to the same macroscopic conclusion: the NCP is experiencing massive permanent storage loss in the order of hundreds of millions to over a billion cubic meters per year. The numerical difference between the two may stem from two aspects: on the one hand, the one-dimensional consolidation theory tends to systematically overestimate storage loss due to its neglect of horizontal displacement and strain; on the other hand, it is also related to the simplifying assumptions in the InSAR-VSM model (e.g., a single aquifer thickness and a defined porosity range). The one-dimensional consolidation theory for estimating GWSL requires explicit separation of the inelastic and elastic components of deformation. However, the primary reason we did not adopt this method in this study lies in the practical challenges of its application at a large regional scale. These are: 1) Accurately parameterizing inelastic compaction requires historical data, such as the preconsolidation head, which are extremely scarce and difficult to obtain across the entire NCP. Our analysis of aquifer properties confirms that the elastic skeletal storage coefficient (S_{ke}) is very low in the main subsidence funnels (Fig. 12d, Fig. S11), indicating that the total subsidence is overwhelmingly dominated by inelastic compaction. 2) Stitching frame data from different orbits (tracks) to construct a spatio-temporally continuous time series is impractical at such a large scale. This process introduces significant residual errors and brings uncertainty to the estimation of cumulative storage loss. Therefore, under these data-scarce conditions, using the total deformation as the primary input for the InSAR-VSM model is a feasible and well-justified strategy, offering a significant advantage over methods that rely on sparse and incomplete groundwater level records.

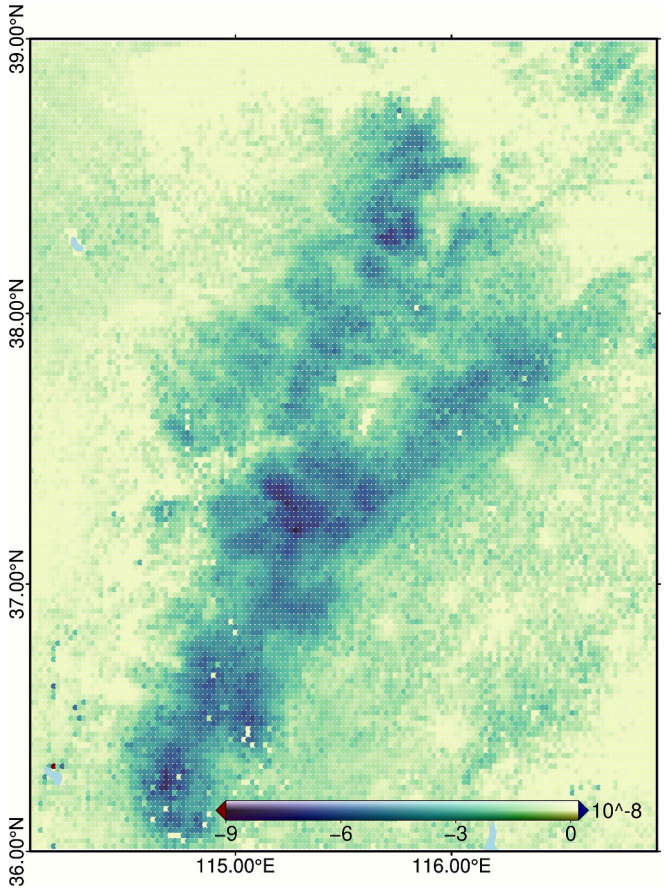

Fig. 13. Volume strain distribution of the groundwater in Zone 1. The location of Zone 1 is shown in Fig. 9a.

However, this study still has the following methodological limitations. 1) Limitations in the InSAR data processing: Although the conversion of satellite LOS deformation to vertical deformation is based on the reasonable assumption that horizontal deformation can be ignored, this conversion may introduce small uncertainties in areas where significant horizontal motion exists locally, such as near fault zones. In addition, the decrease in coherence (Fig.S2) caused by crop growth in the NCP, to some extent, affects the data quality and continuity in some regions. 2) Limitations of the mechanical modeling/parameter inversion: In the InSAR-VSM model, in order to simplify the solution process, we approximate the multi-layered groundwater aquifer as a thicker single-layer aquifer. The actual aquifer system is a complex multi-layered structure, and this simplification may not fully capture the

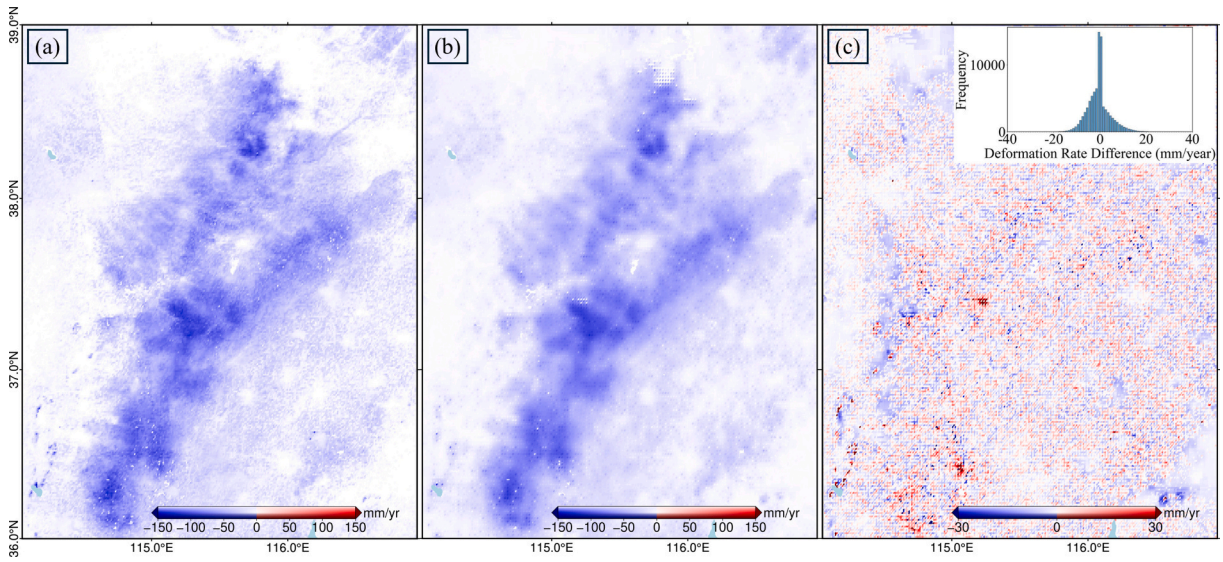

Fig. 14. (a) Downsampled vertical deformation. (b) Estimated vertical deformation. (c) Differences between the estimated and downsampled vertical deformation.

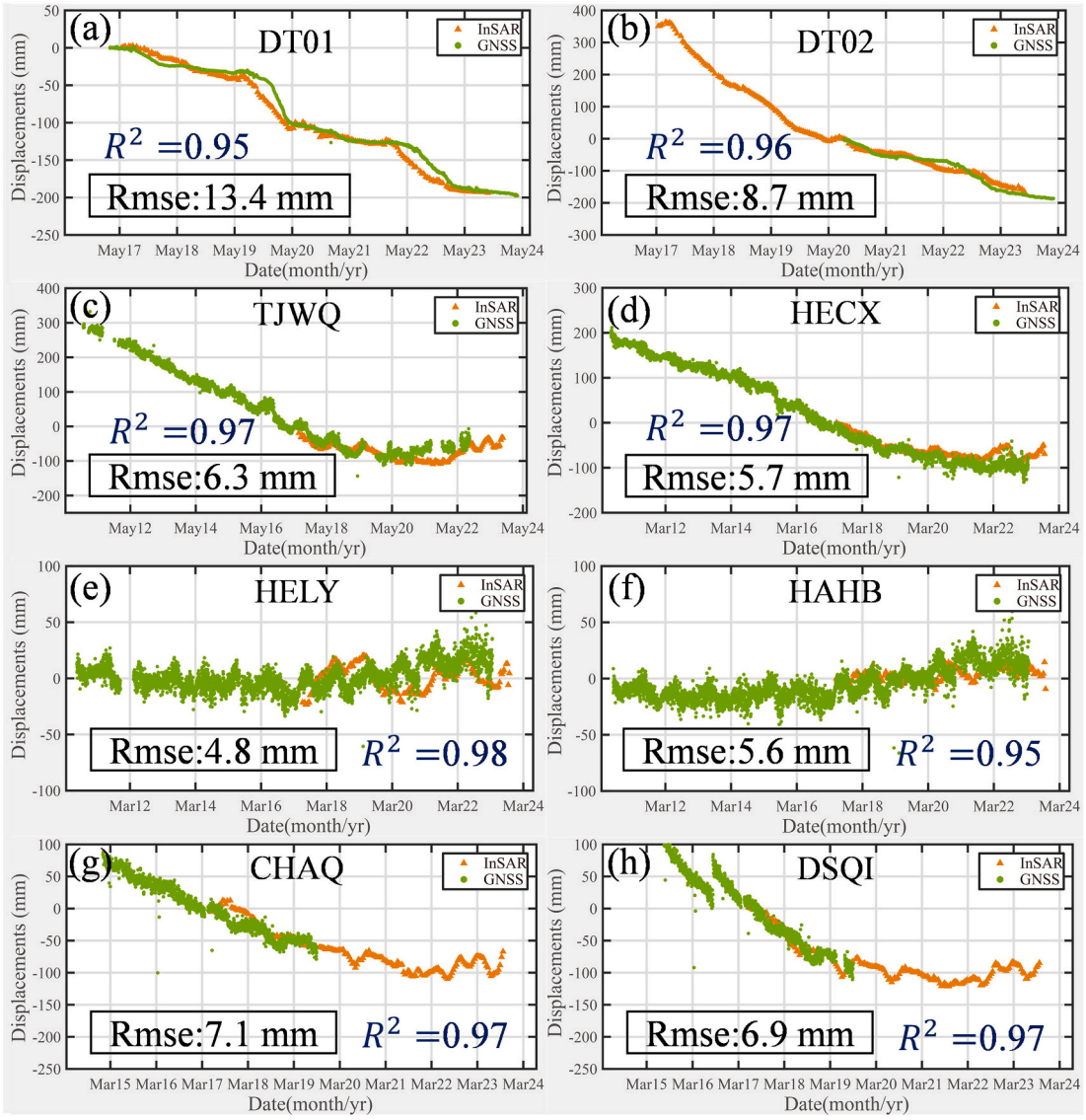

Fig. 15. Comparison between the InSAR and GNSS data, confirming the accuracy of the InSAR results.

hydraulic connections and differentiated responses between aquifers at different depths. For instance, the timing and magnitude of compaction in deeply confined aquifers versus shallower unconfined layers might differ, and a single-layer model averages these responses. This could potentially lead to a generalized representation of the volumetric strain, possibly underestimating or overestimating the compaction in specific layers. The Green's function, which is inherently derived from elastic half-space theory, assumes a homogenous elastic medium, which is an idealization compared to the heterogeneous subsurface of the NCP. While this provides a robust first-order approximation, actual non-elastic behavior or complex layering could introduce discrepancies not fully captured by the current formulation. Meanwhile, the value of the Poisson's ratio (0.25) in the model is based on empirical values obtained from lithological testing of loose Quaternary strata. However, the mechanical parameters of the actual aquifer media may vary with the depth and compaction degree. The assumption of homogenization of these parameters brings inherent uncertainty to the model solution. More critically, the porosity (n) values between aquifers are within a certain range (specifically, we considered n ranging from 0.2 to 0.4 for the GWSL calculation). This range directly impacts the estimated groundwater volume loss. Future research could benefit from more extensive in-situ measurements or geophysical inversions to derive spatially variable and depth-dependent porosity maps, which would improve the GWSL estimation.

In this study, the storage space of the aquifer was discretized into grids with a unit resolution of 2 km × 2 km. Although this resolution represents a great improvement, compared to the traditional large-scale hydrological models and GRACE satellite data (typically with a spatial resolution of > 300 km), the choice of spatial discretization itself strikes a balance between computational feasibility and the ability to capture very fine spatial variations in groundwater loss. Coarse grids may smooth out local features, while finer grids will greatly increase the computational burden and may not provide effective spatial resolution support for InSAR observations in all regions, especially those affected by coherence loss. Our forward modeling validation (Fig. 14) shows the random distribution of the residuals, with a maximum residual of 20 mm, indicating that the selected 2 km × 2 km discretization fully captures the main large-scale volumetric strain patterns. These limitations are inherent challenges for the current research methods, but this study maximizes the reliability of the results through multi-source data integration and rigorous accuracy analysis. Future research will strive to overcome these limitations, in order to gain a more comprehensive and refined understanding.

6.2. The relationship between ground subsidence and faults

The superimposed analysis of faults and surface subsidence helps to further explore the interaction between deformation and faults. Firstly, by overlaying the surface subsidence results of the NCP with the spatial distribution of faults (Fig. 16), it is evident that the spatial distribution of the subsidence funnel is relatively concentrated in a narrow north–south belt, which is consistent with the strike of the faults. The subsidence funnel is surrounded by the Taihang Mountain fault and the BTK fault, indicating that these faults may function as hydraulic barriers (i.e., they block groundwater flow channels and redirect the direction of groundwater flow). However, the influence of active fault zones on groundwater-induced surface deformation is highly complex. For example, near the AY and ND faults, the gradient of the surface deformation is high (Fig. 16 b–d), which may be related to the higher vertical hydraulic conductivity near the faults, thereby increasing the water flow

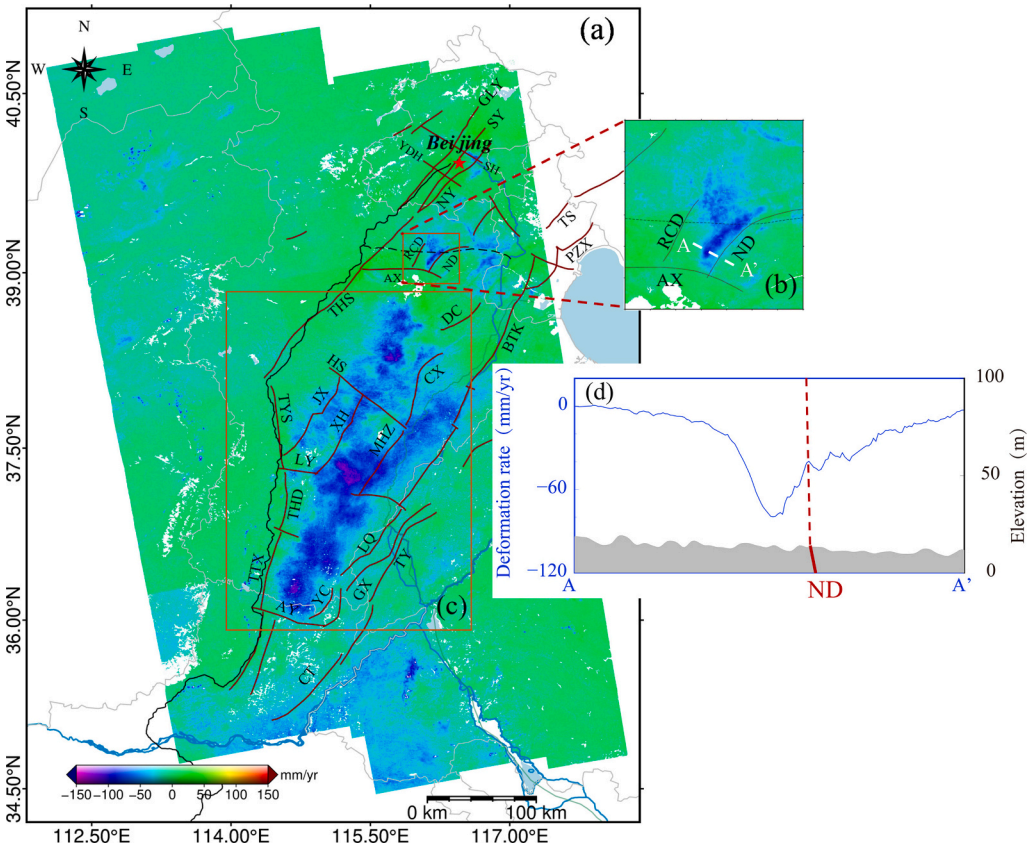

Fig. 16. Analysis of the superposition of faults and surface deformation. (a) The fault distribution is superimposed on the vertical deformation rate map, the dark red lines are faults, and the names of the faults are shown in Table S1. (b) Enlarged map of deformation rates corresponding to the small red box in (a), detailing the deformation field around the RCD, ND, and AX faults. (c) The large red box indicates the main deformation area shown in(a). (d) Vertical deformation rates and elevation along the AA' profile. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

along the faults. It is worth noting that the high vertical hydraulic conductivity of the AY and ND faults does not affect their low horizontal hydraulic conductivity. Therefore, the AY and ND faults act both as barriers to cross-fault fluid flow and as pathways for groundwater flow along the faults.

There are other faults within the NCP that do not substantially hinder surface subsidence expansion, suggesting that the deformation effect of faults in localized areas of the NCP may vary due to the differences in structural features or clay content. Overall, the complex geological structure is the primary controlling factor for the complexity of the surface deformation in the NCP. This complex geological control mechanism highlights the necessity for a deeper understanding of the aquifer system's stress-strain behavior within the regional tectonic context, providing a basis for precise groundwater management in the future.

6.3. *The relationship between ground subsidence and heavy rainfall*

In recent years, driven by global warming, the climate in northern China has trended toward warmer and more humid conditions, accompanied by an increased frequency of intense rainstorms during the summer months. On 19 July 2021 and 29 July 2023, the NCP experienced two once-in-a-century heavy rainfall events, referred to as the "20210720" and "20230729" rainfall events. We investigated the impact of these heavy rainfall events on ground subsidence in the NCP. Firstly, we used Sentinel SAR data to detect and extract flood changes during the "20210720" and "20230729" heavy rainfall events, and performed an overlap analysis with ground deformation, as shown in Fig. 17a–b. The red areas represent the flooded regions, primarily along rivers, and converge in areas with significant subsidence, which further suggests that subsidence-prone regions are more susceptible to flooding disasters.

To further investigate the temporal response between heavy rainfall and ground subsidence, we analyzed the vertical time-series deformation at three representative points (P1, P2, and P3) located in different subsidence bowls (Fig. 17a), as shown in Fig. 18c. Although these points exhibit different long-term subsidence rates, they all show a highly consistent response pattern during the "20,210,720" and "20,230,729" heavy rainfall events: land subsidence significantly decelerated and even experienced a temporary rebound.

To uncover the physical mechanism behind this surface response, we incorporated groundwater level data from monitoring wells within the flood plains (Fig. 18a). The data reveal a rapid and widespread rise in groundwater levels immediately following the "20,210,720" heavy rainfall event, with an average increase of 7.23 m from the heavy rainfall time to the maximum amplitude of the groundwater level (Fig. 18b). This significant rise in groundwater level directly corresponds to a substantial, albeit temporary, recharge of groundwater storage (i.e., a short-term volume gain), thus providing the physical basis for the observed temporary surface rebound. However, this mitigating effect is transient. After reaching its peak, the regional average groundwater level quickly returned to its previous seasonal groundwater level, and

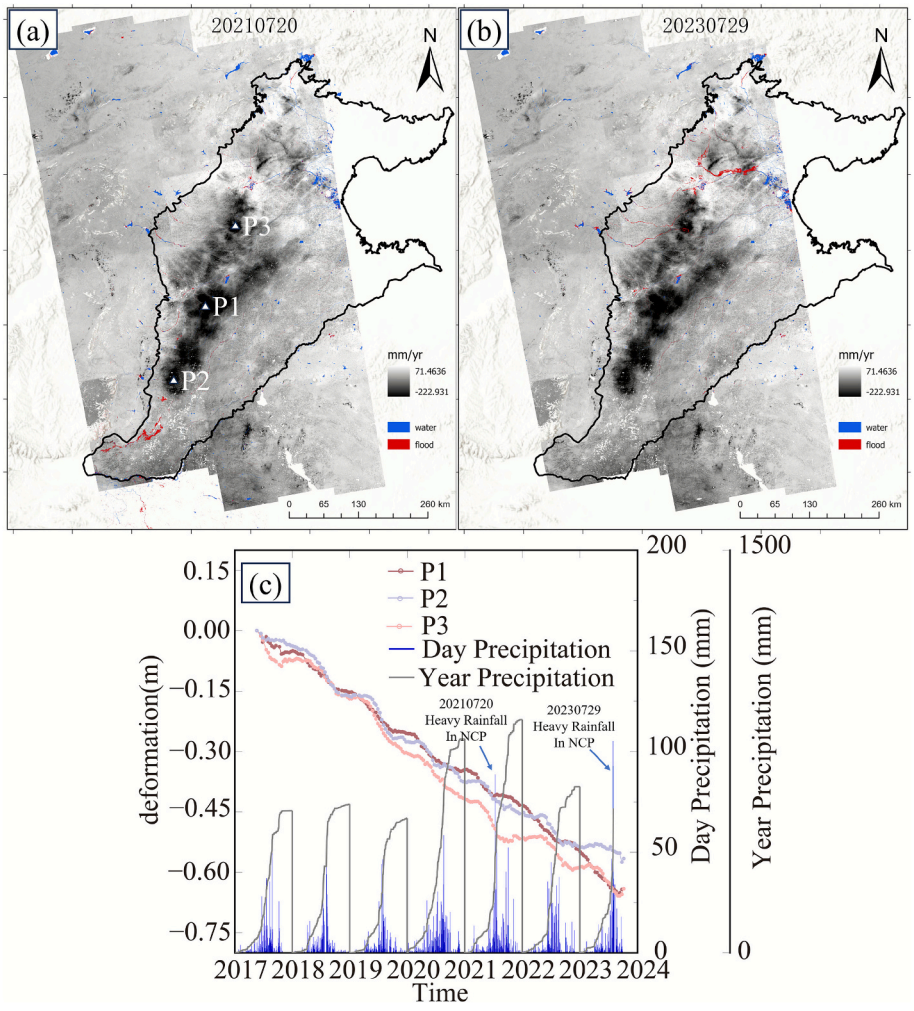

Fig. 17. Analysis of the superposition of extreme rainfall and surface deformation. (a) "20,210,720" flood distribution. (b) "20,230,729" flood distribution. (c) Quantitative assessment of the correlation between the vertical ground displacement and rainfall time series.

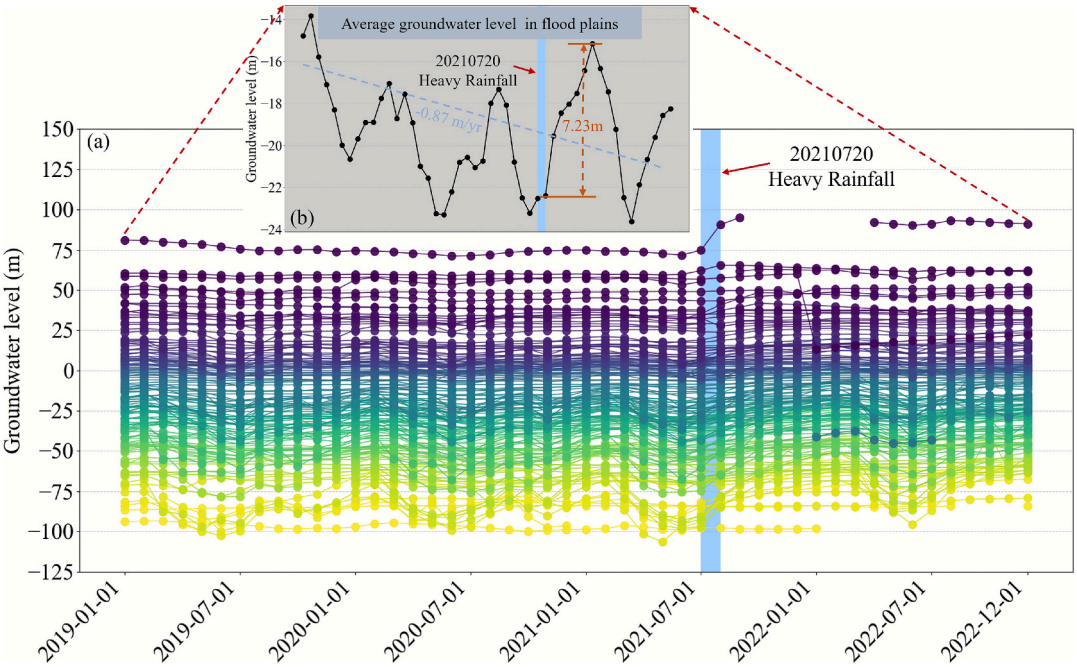

Fig. 18. Groundwater level response to the "20,210,720" heavy rainfall event. (a) Time series of groundwater levels from multiple monitoring wells within the flood plain between 2019 and 2022. Each colored line represents an individual well, and the light blue vertical bar highlights the rainfall period. (b) The average groundwater level time series for wells within the flood plain. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

correspondingly, land subsidence resumed. Thus, we conclude that while extreme rainfall can temporarily mitigate ground subsidence via rapid aquifer recharge, it is insufficient to fundamentally reverse the long-term subsidence trend, which is characterized by a groundwater decline of 0.87 m/yr between 2019 and 2022 and is primarily driven by human activities. This phenomenon is not unique to the NCP. A similar response pattern of temporary rainfall-induced rebound failing to offset long-term subsidence has also been documented in other major aquifer systems, such as California's Central Valley (Murray and Lohman, 2018). This further emphasizes that inelastic aquifer deformation caused by over-extraction is a key factor leading to irreversible subsidence in the NCP, and natural recharge alone is insufficient to fundamentally reverse this trend.

6.4. *Temporal and spatial changes in surface subsidence in the North China plain over 63 years*

By comparing historical subsidence data recorded through traditional leveling surveys with GNSS and InSAR measurement results, we can better understand the long-term dynamics of land subsidence, encompassing the transition process from initial subsidence to cessation and subsequent ground rebound. Firstly, we obtained leveling survey results for the NCP in the following three periods: 1960–1980, 1980–1990, and 1990–2010 (Fig. S12). Based on these three-phase leveling results, we further plotted cumulative land subsidence contour maps for 1960–2010 (Fig. 19a). The 2017–2023 deformation velocity map was derived from the Sentinel datasets (Fig. 19b). According to the data leveling, the considerable land subsidence between 1960 and 1980 primarily occurred in urban areas such as Tianjin and Cangzhou, while farmland areas in the NCP showed slow subsidence development. From 1980 to 1990, rapid economic growth led to increased groundwater extraction, accelerating subsidence expansion, with emerging subsidence in agricultural regions. From 1990 to 2010, farmland subsidence intensified dramatically, while deformation in the Tianjin and Cangzhou urban areas showed relative deceleration. By 2010, the maximum cumulative subsidence in Tianjin's subsidence center had reached nearly 3 m, while the Cangzhou subsidence center recorded 1.8 m. Multiple subsidence funnel centers emerged in farmland areas, with a maximum cumulative subsidence of up to 2 m.

Historical subsidence occurred primarily in urban areas or regions with the deep groundwater level decline exceeding 70 m, due to over-exploitation. Current subsidence is observed in the piedmont plain areas with shallow groundwater systems, with a 40-m head decline (Fig. 19a–b). We further obtained time-series deformation results for Tianjin from 1960 to 2023 (Fig. 19c), integrating leveling data with the time-series GNSS and InSAR data. Land subsidence in Tianjin followed a linear trend from 1960 to 2019, stabilized during 2019–2020, and exhibited a rebound phenomenon after May 2020 (Fig. 19c). Over this 63-year period, land subsidence in Tianjin has undergone complete phases of accelerated subsidence, deceleration, and eventual ground uplift. Through analyzing the spatiotemporal evolution patterns in the NCP, we conclude that: 1) the deformation center has shifted from coastal urban areas to piedmont farmland regions; and 2) coastal cities show stabilized or rebounding deformation trends, while farmland subsidence continues to intensify. This spatial migration characteristic and the varying deformation trends reflect a profound shift in driving mechanisms. The mitigation or rebound of subsidence in urban areas may be closely related to the SNWDP, which provides alternative water sources, reduces groundwater extraction, and implements related policies prohibiting groundwater extraction. In contrast, the sustained intensification of subsidence in agricultural areas can be primarily attributed to their heavy reliance on deep groundwater for irrigation. This indicates that, despite overall measures having been taken, the implementation and effectiveness of the water resource management policies vary significantly across different regions, necessitating more refined management strategies.

6.5. *The role of the SNWDP in multi-factor subsidence mitigation in the North China plain*

The impact of the SNWDP on groundwater storage and land deformation in the NCP can be reflected through the temporal deformation

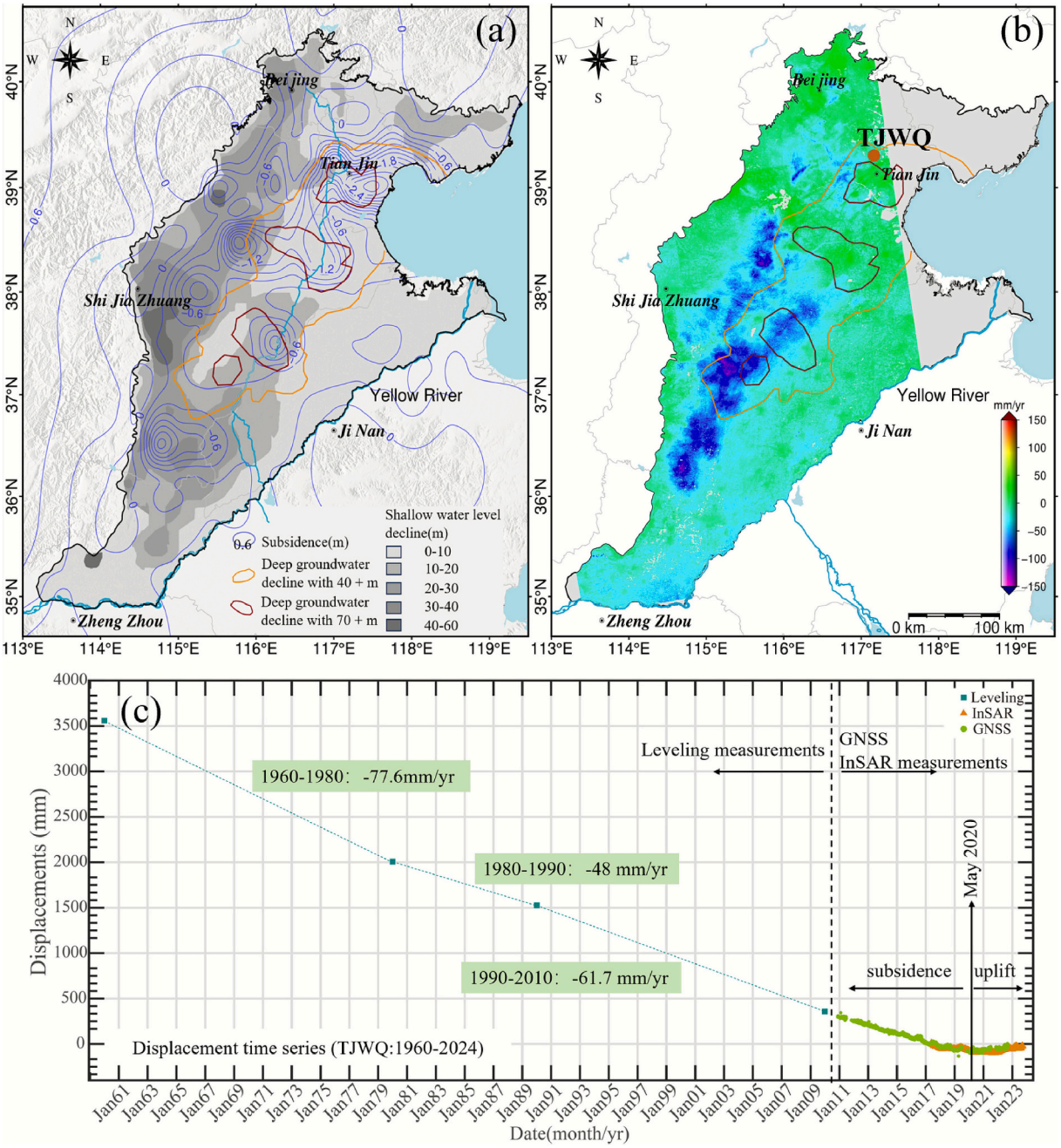

Fig. 19. Temporal and spatial changes of surface subsidence in the NCP over 63 years. (a) Land subsidence was observed cumulatively from the 1960s to the 2010s, with corresponding declines in shallow (1970–2008) and deep (1970–2003) groundwater levels. (b) The InSAR results from 2017 to 2023. (c) The 63-year cumulative deformation map (TJWQ) was generated by integrating leveling, GNSS, and InSAR datasets.

changes in the region. To further investigate its temporal evolution, an exponential function $def(t) = M(e^{kt} - 1)$ was employed (Hu and Bürgmann, 2020).$def(t)$ is the time-series vertical deformation, M is the cumulative displacement, and $k \in [-1, 0]$. The closer the k value is to -1, the faster the deformation rate decreases. Conversely, the closer the k value is to 0, the slower the deformation rate decreases. The best-fit values for k and M were obtained using a non-linear least-squares fitting algorithm for each time-series InSAR point. We selected 11 major cities in the NCP with representative subsidence patterns for analysis. As shown in Fig. 20, the subsidence trends across the NCP exhibit significant regional variation. We divided the region into two parts, using the line from Baoding to Cangzhou as the boundary. The upper part of the region shows a decreasing deformation trend, with cities such as Beijing and Tianjin serving as typical examples (Fig. 20 b–d, f). This success reflects the synergistic effects of a "dual strategy", i.e., the provision of alternative water sources by the SNWDP combined with stringent local groundwater management policies. The SNWDP has acted as a critical enabler, creating the necessary conditions for groundwater recovery, while strong local governance has ensured its effectiveness.In contrast, the lower part of the region shows a continuous increasing deformation trend, with cities such as Shijiazhuang, Xingtai, and Handan as typical examples (Fig. 20 e, g–m). Although the middle route of the SNWDP diverts up to 9.5 km^3 of water annually (Long et al., 2020), there is a clear disparity in its allocation, as shown in Table 3. The majority of this water is allocated for industrial use in large cities such as Beijing and Tianjin. In contrast, for the vast agricultural areas, which are predominantly in Hebei province, the water resource allocation density is only

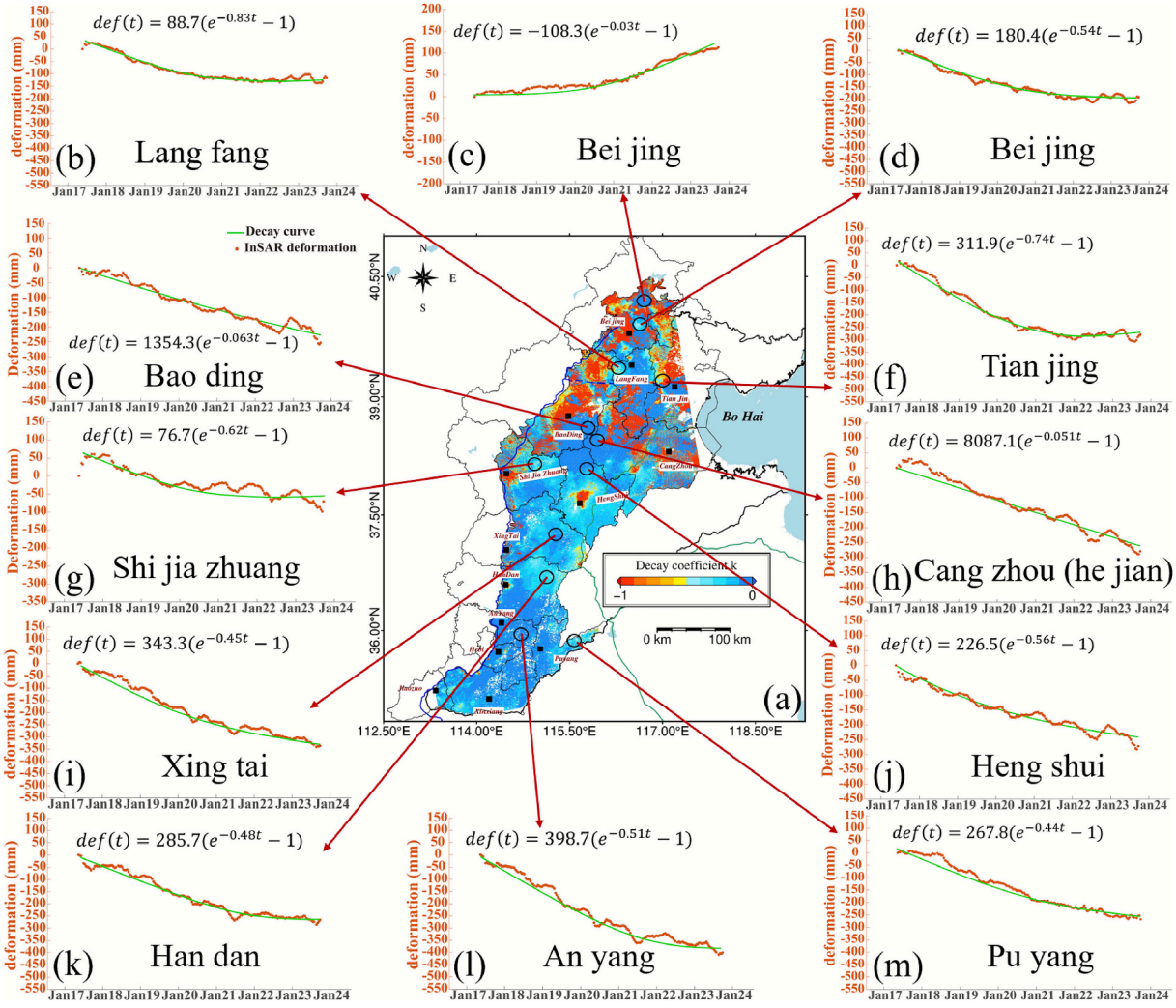

Fig. 20. Deformation trend distribution in the NCP and deformation modeling of typical cities. (a) Spatial distribution of the decay coefficient (k) from the fitted exponential model, which indicates the rate of subsidence deceleration in the NCP. (b)–(m) Comparison of the time-series InSAR data and exponential decay model fitting for 12 typical city areas. The black circles mark the locations of the 12 typical cities in Fig. 20a.

Table 3
Annual water allocation data (2017–2022) from the middle route of the SNWDP, sourced from the Ministry of Water Resources of China and apportioned to the city level based on the allocation ratios from Zhang et al. (2021).

	Province	City	City area (10^3 km^2)	Province area (10^3 km^2)	Allocation ratio at the province level (%)	Water resource allocation density per unit area	Allocation ratio at the city level (%)	Water supply (billion cubic meters)					
								2017	2018	2019	2020	2021	2022
NCP	Beijing	Beijing	16.4	16.4	24.8	1.51	24.8	12.02	18.50	17.69	21.38	20.99	22.85
	Tianjin	Tianjin	11.9	11.9	20.0	1.68	20.0	9.70	14.92	14.26	17.24	16.93	18.42
	Hebei	Tangshan	13.4	104.6	21.6	0.20	–	–	–	–	–	–	–
		Langfang	6.5				0.2	0.1	0.15	0.14	0.17	0.17	0.18
		Baoding	22.2				5.2	2.52	3.88	3.71	4.48	4.40	4.79
		Cangzhou	13.4				0.1	0.05	0.07	0.07	0.09	0.08	0.09
		Shijiazhuang	15.8				12.0	5.82	8.95	8.56	10.35	10.16	11.05
		Hengshui	8.8				1.1	0.53	0.82	0.78	0.95	0.93	1.01
		Xingtai	12.5				1.4	0.68	1.04	1.00	1.21	1.18	1.29
		Handan	12				1.7	0.82	1.27	1.21	1.47	1.44	1.57
		Anyang	5.6				8.0	3.88	5.97	5.71	6.90	6.77	7.37
		Puyang	4.1				4.3	2.08	3.21	3.07	3.71	3.64	3.96
	Henan	Xinxiang	8.2	24.3	33.6	1.38	9.1	4.41	6.79	6.49	7.85	7.70	8.38
		Hebi	2.3				5.2	4.41	6.79	6.49	7.85	7.70	8.38
		Jiaozuo	4.1				6.9	3.35	5.15	4.92	5.95	5.84	6.36

0.2, which is considerably less than in the other regions. This limited supplemental water is insufficient to meet irrigation demands, forcing these regions to continue their reliance on groundwater. As demonstrated in Fig. 20, subsidence in the NCP's farmland areas continues to intensify.

This situation underscores that the long-term sustainability of groundwater resources and the ultimate halt of subsidence across the NCP depend on the continued integration of large-scale water diversion projects with robust, locally enforced water management policies and targeted agricultural water-saving initiatives. Therefore, optimizing groundwater sustainability requires not only a strategic adjustment of water allocation (shifting more SNWDP water toward agricultural regions), but also tackling the associated practical challenges. This includes investing in "last-mile" infrastructure to efficiently deliver water from main canals to farmland. Furthermore, developing viable water pricing mechanisms and subsidy policies will be crucial to ensure that the diverted water is an economically feasible alternative for farmers. Only when combined with continuous monitoring and adaptive management can these integrated strategies create a sustainable path forward for this vital region.

7. Conclusions

In this paper, we have presented the first comprehensive, high-resolution (2 km) assessment of groundwater storage loss and land subsidence across the entire NCP from 2017 to 2023. By applying a synergistic framework of wide-area InSAR and mechanical modeling, we systematically quantified the complex dynamics of the aquifer system. Our main findings are as follows:

1. This study produced a seamless vertical deformation field for the entire NCP with unprecedented precision. Three severe subsidence areas were identified, with the maximum vertical deformation rate exceeding −150 mm/yr, resulting in a cumulative subsidence of up to 1 m over six years. A comprehensive analysis of data from the past 63 years revealed a significant spatial migration of subsidence centers from coastal urban areas to inland agricultural regions. Furthermore, deformation in urban areas has stabilized or even reversed, while subsidence in agricultural areas continues to intensify.
2. The research reveals a strong coupling relationship between land subsidence and groundwater levels, with correlation coefficients as high as 0.94 to 0.98. Meanwhile, cross-wavelet analysis of data from a floodplain well (W8) quantified the dynamic coupling between groundwater levels and surface deformation. The analysis revealed a dominant one-year resonance period, with a remarkably short two-day lag time for the surface to respond to groundwater changes, confirming the aquifer system's rapid reaction to hydraulic variations. In addition, we quantitatively analyzed the nonlinear response characteristics of the Quaternary aquifer system to 300 groundwater level changes, obtaining S_{ke} for the NCP ranging from 0.0002 to 0.021. We confirmed the different responses of the various hydrogeological units: the piedmont plain is dominated by elastic deformation, whereas the flood plain is characterized by plastic deformation. The latter exhibits almost no elastic recovery capacity, leading to the continuous exacerbation of permanent groundwater storage loss.
3. Quantitative assessment of groundwater loss in the NCP aquifer is a crucial aspect of hydrological research. Based on the InSAR-VSM groundwater volume inversion model, we improved the spatial resolution of the groundwater storage inversion in the NCP from hundreds of kilometers to 2 km for the first time. We quantified the average groundwater loss from 2017 to 2023 to be -4.346×10^8 to -8.692×10^8 m^3/yr. This research has broken through the bottleneck of insufficient spatial resolution in groundwater hydrological studies, effectively compensating for the lack of ground observation data and the difficulty of estimating detailed hydrological parameters using GRACE satellite data at a hundred-kilometer scale.
4. This study systematically investigated the complex evolution of land subsidence under the combined influence of human activities (e.g., excessive groundwater extraction, the SNWDP, and groundwater no-extraction policies) and natural activities (e.g., geological structures, extreme rainfall). It was confirmed that the active faults influence the spatial distribution of the subsidence funnels. While extreme rainfall events can cause a temporary slowdown of subsidence, they are unable to reverse the long-term trend driven by over-extraction. The SNWDP has played a positive role in mitigating subsidence in urban areas such as Beijing and Tianjin, but its limited allocation to agricultural areas has failed to halt the intensifying subsidence in these areas.

The collaborative analysis framework of comprehensive remote sensing monitoring and mechanical modeling revealed the spatiotemporal evolution laws of deformation of the aquifer systems and groundwater reserves, providing a new understanding of the stress evolution of aquifer systems at the watershed scale, and offering critical scientific insights for adaptive groundwater management and the mitigation and control of land subsidence in water-deficient sedimentary basins.

CRediT authorship contribution statement

Xing Zhang: Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Formal analysis. **Jun Hu:** Writing – review & editing, Supervision, Software, Resources, Funding acquisition. **Mahdi Motagh:** Writing – review & editing, Methodology, Conceptualization. **Mingjia Li:** Writing – review & editing, Resources. **Yuedong Wang:** Writing – review & editing, Methodology. **Qiuhong Yang:** Writing – review & editing, Resources. **Guangli Su:** Writing – review & editing, Funding acquisition. **Haigang Wang:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the European Space Agency for providing the Sentinel-1 SAR data freely. The authors would also like to thank the China Geological Environment Monitoring Institute Geo-Cloud (https://geocloud.cgs.gov.cn) for providing the groundwater data. The authors would like to thank Dr. Qing Yang for providing the flood change detection results. This work was supported by the National Natural Science Foundation of China (Nos. 42030112, 42474033, 42204008, and 42404022.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2025.115164.

Data availability

Data will be made available on request.

References

Ao, Z., Hu, X., Tao, S., Hu, X., Wang, G., Li, M., Wang, F., Hu, L., Liang, X., Xiao, J., 2024. A national-scale assessment of land subsidence in China's major cities. Science 384 (6693), 301–306.

Aster, R.C., Borchers, B., Thurber, C.H., 2018. Parameter Estimation and Inverse Problems. Elsevier.

Bai, L., Jiang, L., Zhao, Y., Li, Z., Cao, G., Zhao, C., Liu, R., Wang, H., 2022. Quantifying the influence of long-term overexploitation on deep groundwater resources across Cangzhou in the North China plain using InSAR measurements. J. Hydrol. 605. https://doi.org/10.1016/j.jhydrol.2021.127368.

Berardino, P., Fornaro, G., Lanari, R., Sansosti, E., 2002. A new algorithm for surface deformation monitoring based on small baseline differential SAR Interferograms. IEEE Transact. Geosci. Remote Sens. 40 (11), 2375–2383.

Chaussard, E., Farr, T.G., 2019. A new method for isolating elastic from inelastic deformation in aquifer systems: application to the San Joaquin Valley, CA. Geophys. Res. Lett. 46 (19), 10800–10809. https://doi.org/10.1029/2019gl084418.

Cigna, F., Tapete, D., 2021. Satellite InSAR survey of structurally-controlled land subsidence due to groundwater exploitation in the Aguascalientes Valley, Mexico. Remote Sens. Environ. 254. https://doi.org/10.1016/j.rse.2020.112254.

Deng, Q.D., Zhang, P.Z., Ran, Y.K., Yang, X.P., Wei, M., Chen, L.C., 2003. Active tectonics and earthquake activities in China (in Chinese). Earth Sci. Front. 10 (1), 66.

Dong, J., Guo, S., Wang, N., Zhang, L., Ge, D., Liao, M., Gong, J., 2023. Tri-decadal evolution of land subsidence in the Beijing plain revealed by multi-epoch satellite InSAR observations. Remote Sens. Environ. 286, 113446.

Dong, L., Long, D., Zhang, C., Cui, Y., Cui, Y., Wang, Y., Li, L., Hong, Z., Yao, L., Quan, J., 2024. Shifting agricultural land use and its unintended water consumption in the North China plain. Sci. Bull. 69 (24), 3968–3977.

Garg, S., Motagh, M., Indu, J., Karanam, V., 2022. Tracking hidden crisis in India's capital from space: implications of unsustainable groundwater use. Sci. Rep. 12 (1), 651.

Ge, D.Q., Wang, Y., Zhang, L., Xia, Y., Guo, X.F., 2010. Large Scale Land Subsidence Monitoring with a Reduced Set of SAR Images. IEEE.

Gong, H., Pan, Y., Zheng, L., Li, X., Zhou, C., 2018. Long-term groundwater storage changes and land subsidence development in the North China plain (1971–2015). Hydrogeol. J. 1, 1–11.

Haghshenas Haghighi, M., Motagh, M., 2024. Uncovering the impacts of depleting aquifers: a remote sensing analysis of land subsidence in Iran. Sci. Adv. 10 (19), eadk3039.

Hasan, M.F., Smith, R., Vajedian, S., Pommerenke, R., Majumdar, S., 2023. Global land subsidence mapping reveals widespread loss of aquifer storage capacity. Nat. Commun. 14 (1), 6180. https://doi.org/10.1038/s41467-023-41933-z.

Herrera-Garcia, G., Ezquerro, P., Tomas, R., Bejar-Pizarro, M., Lopez-Vinielles, J., Rossi, M., Mateos, R.M., Carreon-Freyre, D., Lambert, J., Teatini, P., Cabral-Cano, E., Erkens, G., Galloway, D., Hung, W.C., Kakar, N., Sneed, M., Tosi, L., Wang, H., Ye, S., 2021. Mapping the global threat of land subsidence. Science 371 (6524), 34–36. https://doi.org/10.1126/science.abb8549.

Hu, X., Bürgmann, R., 2020. Aquifer deformation and active faulting in salt Lake Valley, Utah, USA. Earth Planet. Sci. Lett. 547. https://doi.org/10.1016/j.epsl.2020.116471.

Jasechko, S., Seybold, H., Perrone, D., Fan, Y., Shamsudduha, M., Taylor, R.G., Fallatah, O., Kirchner, J.W., 2024. Rapid groundwater decline and some cases of recovery in aquifers globally. Nature 625 (7996), 715–721.

Kakar, N., Metzger, S., Schöne, T., Motagh, M., Waizy, H., Nasrat, N., Lazecký, M., Amelung, F., Bookhagen, B., 2025. Interferometric radar satellite and in-situ well time-series reveal groundwater extraction rate changes in urban and rural Afghanistan. Water Resour. Res. 61 (3), e2023WR036626.

Kalia, A.C., Frei, M., Lege, T., 2017. A Copernicus downstream-service for the nationwide monitoring of surface displacements in Germany. Remote Sens. Environ. 202, 234–249.

Long, D., Yang, W., Scanlon, B.R., Zhao, J., Liu, D., Burek, P., Pan, Y., You, L., Wada, Y., 2020. South-to-north water diversion stabilizing Beijing's groundwater levels. Nat. Commun. 11 (1), 3665. https://doi.org/10.1038/s41467-020-17428-6.

Motagh, M., Shamshiri, R., Haghighi, M.H., Wetzel, H.-U., Akbari, B., Nahavandchi, H., Roessner, S., Arabi, S., 2017. Quantifying groundwater exploitation induced subsidence in the Rafsanjan plain, southeastern Iran, using InSAR time-series and in situ measurements. Eng. Geol. 218, 134–151.

Motagh, M., Garg, S., Cigna, F., Teatini, P., Bhardwaj, A., Matin, M.A., Zarei, A., Madani, K., 2024. Sustainability Nexus AID: Landslides and Land Subsidence, Paper Presented at Sustainability Nexus Forum. Springer.

Murray, K.D., Lohman, R.B., 2018. Short-lived pause in Central California subsidence after heavy winter precipitation of 2017. Sci. Adv. 4 (8), eaar8144.

Neely, W.R., Borsa, A.A., Burney, J.A., Levy, M.C., Silverii, F., Sneed, M., 2021. Characterization of groundwater recharge and flow in California's San Joaquin Valley from InSAR-observed surface deformation. Water Resour. Res. 57 (4). https://doi.org/10.1029/2020WR028451 e2020WR028451.

Payne, J.A., Watson, A., Maghsoudi, Y., Ebmeier, S., Rigby, R., Lazecký, M., Thomas, M., Elliott, J., 2025. Widespread extent of irrecoverable aquifer depletion revealed by country-wide analysis of land surface subsidence hazard in Iran. J. Geophys. Res. Solid Earth 130 (9) e2024JB030367.

Peng, J., Xu, J., Ma, R., Wang, F., 2016. Characteristics and mechanism of the Longyao ground fissure on North China plain, China. Eng. Geol. 214, 136–146.

Peng, M., Lu, Z., Zhao, C., Motagh, M., Bai, L., Conway, B.D., Chen, H., 2022. Mapping land subsidence and aquifer system properties of the Willcox Basin, Arizona, from InSAR observations and independent component analysis. Remote Sens. Environ. 271. https://doi.org/10.1016/j.rse.2022.112894.

Rodell, M., Famiglietti, J.S., Wiese, D.N., Reager, J., Beaudoing, H.K., Landerer, F.W., Lo, M.-H., 2018. Emerging trends in global freshwater availability. Nature 557 (7707), 651–659.

Shi, M., Gong, H., Gao, M., Chen, B., Zhou, C., 2020. Recent ground subsidence in the North China plain, China, revealed by sentinel-1A datasets. Remote Sens. 12 (21), 3579.

Smith, R., Li, J., 2021. Modeling elastic and inelastic pumping-induced deformation with incomplete water level records in Parowan Valley, Utah. J. Hydrol. 601. https://doi.org/10.1016/j.jhydrol.2021.126654.

Steinshouer, D.W., Qiang, J., Mccabe, P.J., Ryder, R.T., 1997. Maps Showing Geology, Oil and Gas Fields, and Geologic Provinces of the Asia Pacific Region.

Su, G., Xiong, C., Zhang, G., Wang, Y., Shen, Q., Chen, X., An, H., Qin, L., 2023. Coupled processes of groundwater dynamics and land subsidence in the context of active human intervention, a case in Tianjin, China. Sci. Total Environ. 903, 166803.

Tang, W., Zhao, X., Motagh, M., Bi, G., Li, J., Chen, M., Chen, H., Liao, M., 2022. Land subsidence and rebound in the Taiyuan basin, northern China, in the context of inter-basin water transfer and groundwater management. Remote Sens. Environ. 269. https://doi.org/10.1016/j.rse.2021.112792.

Wang, S., Lu, Z., Wang, B., Niu, Y., Song, C., Li, X., Ma, Z., Xu, C., 2023. A phase-based InSAR tropospheric correction method for interseismic deformation based on short-period interferograms. IEEE Transac. Geosci. Remote Sens. 61, 1–18.

Wang, Y., Feng, G., Li, Z., Yang, Z., Wang, B., Wang, Y., Du, Y., Wang, Y., He, L., Zhu, J., 2024. A multi-frame deformation velocity splicing method for wide-area InSAR measurement based on uncontrolled block adjustment: a case study of long-term deformation monitoring in Guangdong, China. Remote Sens. Environ. 301, 113929.

Yague-Martinez, N., Prats-Iraola, P., Rodriguez Gonzalez, F., Brcic, R., Shau, R., Geudtner, D., Eineder, M., Bamler, R., 2016. Interferometric processing of Sentinel-1 TOPS data. IEEE Trans. Geosci. Remote Sens. 54 (4), 2220–2234. https://doi.org/10.1109/tgrs.2015.2497902.

Zhang, C., Duan, Q., Yeh, P.J.-F., Pan, Y., Gong, H., Moradkhani, H., Gong, W., Lei, X., Liao, W., Xu, L., 2021. Sub-regional groundwater storage recovery in North China plain after the south-to-north water diversion project. J. Hydrol. 597, 126156.

Zhang, X., Ren, L., Feng, W., 2022. Comparison of the shallow groundwater storage change estimated by a distributed hydrological model and GRACE satellite gravimetry in a well-irrigated plain of the Haihe River basin, China. J. Hydrol. 610. https://doi.org/10.1016/j.jhydrol.2022.127799.

Zhang, X., Hu, J., Jin, X., Liu, X., Liu, J., Zheng, W., Wu, W., Duan, M., 2024b. Resolving Three-Dimensional Displacements Caused by Subsurface Fluid Activities From InSAR and Volumetric Strain Model: Adaptive Estimation of Model Parameters. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 17, 4859–4874.

Zhang, X., Sun, Q., Gui, R., Hu, J., Yang, Q., Su, G., 2025. Long-time surface deformation in Beijing, China for three decades by multi-sensor, multi-track and multi-temporal InSAR seamless connection. Geomat. Nat. Haz. Risk 16 (1), 2478950.

Zhao, Q., Zhang, B., Yao, Y., Wu, W., Meng, G., Chen, Q., 2019. Geodetic and hydrological measurements reveal the recent acceleration of groundwater depletion in North China plain. J. Hydrol. 575, 1065–1072.

Zhao, X., Yu, Y., Cheng, J., Ding, K., Luo, Y., Zheng, K., Xian, Y., Lin, Y., 2024. A novel framework for heterogeneity decomposition and mechanism inference in spatiotemporal evolution of groundwater storage: case study in the North China plain. Water Resour. Res. 60 (12) e2023WR036102.